SXSW SENTIMENT ANALYSIS PROJECT

Project Summary

This project applies the Data Science NLP methodology to analyze sentiment in tweets related to product and brand mentions.

- **Business Understanding**: Identify how users express emotions toward brands and products on social media.
- **Data Preparation**: Cleaned and preprocessed tweet data, including tokenization, stopword removal, and sentiment labeling.
- **Modeling**: Implemented machine learning models (e.g., Logistic Regression, Random Forest, and Neural Networks) to classify sentiment.
- **Evaluation**: Compared model performance using accuracy, precision, recall, and F1-score.
- **Deployment Insight**: Results highlight how concise yet informative messaging aligns with natural user behavior.

🗁 Deliverables

- Presentation Slides
- Dataset
- Sources

🗀 Repository Structure

- `data/` → raw and processed datasets
- `notebooks/` → Jupyter notebooks for preprocessing, modeling, and evaluation
- `presentation/` → stakeholder slides (PDF/PowerPoint)
- `README.md` → project summary and navigation instructions

How to Navigate & Reproduce

To clone the Repository `bash git clone https://github.com/LilianToo/NLP-Sentiment-Analysis-Project.git cd nlp-sentiment-analysis-project

Overview

In this project, we studied how people feel about different companies and products during SXSW by analyzing tweets from a dataset "judge-1377884607_tweet_product_company". Each tweet includes the company, product category and corresponding sentiment labels. This makes it useful for understanding customer perception and brand performance.

Business understanding

Nowadays, customers express their satisfaction and dissatisfaction on social media especially on platforms like Twitter. The tweets reveal how people feel about brands, products and their customer experience. Companies are increasingly relying on sentiment analysis to monitor public perception, identify dissatisfaction early, and guide marketing, product design, and support decision-making.

Problem Statement

Product owners and companies invest heavily on marketing at major events like SXSW,but lack real-time, data-driven understanding of public opinion. They struggle to answer:

1. Is our campaign resonating positively?
2. How sentiments towards our brand compare to our competitors?
3. What specific product features or marketing stunts are people talking about(Positive/Negative/Neutral)?

Stakeholders

The stakeholders in this project are: 1.Marketing Directors:- They decide whether campaigns are resonating. 2.Brand Consultants:- Advise clients on positioning and trends. 3.Company Managers:- Plan budgets and strategies for future events.

Data Understanding

From the dataset "judge-1377884607_tweet_product_company", each row contains the tweet text,the company referenced, the product category, and whether the sentiment is Positive, Negative or Neutral.The dataset is passed throught the following steps:

Steps

We prepared the data by using the following:

1. Data cleaning
2. Text preprocessing
3. Exploratory Analysis
4. Feature Engineering
5. Balancing Data using SMOTE
6. Modeling
7. Evaluation
8. Visualization.

Data Preparation

This step involves handling the missing values,cleaning the text through lowercasing, removing punctuation, and stopwords to reduce noise and standardize the text for better vectorization. The cleaned text is then tokenized and lemmatized to reduce vocabulary size by converting words to their base forms.

The encoded target labels converted categorical emotion labels into numerical format for modeling.

Stratified train-test split is performed to preserve class balance in both training and testing sets.

Applied TF-IDF vectorization to transform text into meaningful numerical features for ML models.

SMOTE is applied to balance minority classes for improved model performance on underrepresented labels.

Features are scaled to improve convergence and stability for the neural network model.

This enabled us to identify key factors that influence public opinion on different companies and products.

Our analysis revealed key insights such as:

1. Positive sentiment dominates
2. Negative sentiments
3. Tweet length trends
4. Competitive positioning

We conclude with three actionable recommendations based on sentiment patterns that can help companies improve brand reputation and strengthen customer satisfaction.

Strategic Offering

The analysis provides us quantified brand sentiment and competitive intelligence that can help companies make evidence-based decisions in marketing, product design and customer experience.

Data Modeling

We transformed text into features that machine learning models can understand, i.e TF-IDF vectors or embeddings and selecting models capable of learning patterns that indicates the emotion behind each tweet. Models used:

1.Logistic Regression (Baseline Model)

This is a simple and interpretable linear model that works well with text represented as TF-IDF or word counts.

2.Random Forest

It is an ensemble of decision trees and good at handling complex, non-linear data provides feature importance and resistant to overfitting as well as noise.

3. Neural Networks

Flexible and powerful for modeling non-linear relationships. Can combine multiple feature types(embeddings + TF-IDF) This model learns word interactions automatically, often more accurately.

4. BERT

This is a pre-trained deep learning model specialized in language understanding. Particularly effective for short, informal text like tweets, improving emotion classification by leveraging semantic understanding beyond simple word counts. Can be fine-tuned on your dataset for state-of-the-art performance in classification tasks

This enabled us to identify key factors that influence public opinion on different companies and products.

Key insights

Our analysis revealed key insights such as:

Positive sentiment dominates Negative sentiments Tweet length trends Competitive positioning

We conclude with three actionable recommendations based on sentiment patterns that can help companies improve brand reputation and strengthen customer satisfaction.

Strategic Offering

The analysis provides us quantified brand sentiment and competitive intelligence that can help companies make evidence-based decisions in marketing, product design and customer experience.

Importation of Necessary Libraries

In [1]:
```
# Libraries
!pip install tensorflow
!pip install imblearn
!pip install keras-
!pip install plotly
!pip install transformers datasets accelerate torch
```

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Requirement already satisfied: tensorflow in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (2.20.0)
Requirement already satisfied: absl-py>=1.0.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow) (2.3.1)
Requirement already satisfied: astunparse>=1.6.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow) (25.2.10)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow) (0.4.0)
Requirement already satisfied: google_pasta>=0.1.1 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow) (0.2.0)
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Requirement already satisfied: protobuf>=5.28.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow) (6.33.6)
Requirement already satisfied: requests<3,>=2.21.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow) (2.32.5)
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Requirement already satisfied: keras>=3.10.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow) (3.10.0)
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Requirement already satisfied: six>=1.5 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
```

In [2]:
```
!pip install tf-keras
```

```
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Requirement already satisfied: tensorflow<2.21,>=2.20 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tf-keras) (2.20.0)
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Requirement already satisfied: astunparse>=1.6.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (1.6.3)
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Requirement already satisfied: requests<3,>=2.21.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (2.32.5)
Requirement already satisfied: setuptools in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (80.9.0)
Requirement already satisfied: six>=1.12.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (3.1.0)
Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (4.14.1)
Requirement already satisfied: wrapt>=1.11.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (1.17.3)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (1.80.0)
Requirement already satisfied: tensorboard~=2.20.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (2.20.0)
Requirement already satisfied: keras>=3.10.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (3.10.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (1.26.4)
Requirement already satisfied: h5py>=3.11.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (3.14.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorflow<2.21,>=2.20->tf-keras) (0.5.4)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from requests<3,>=2.21.0->tensorflow<2.21,>=2.20->tf-keras) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from requests<3,>=2.21.0->tensorflow<2.21,>=2.20->tf-keras) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from requests<3,>=2.21.0->tensorflow<2.21,>=2.20->tf-keras) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from requests<3,>=2.21.0->tensorflow<2.21,>=2.20->tf-keras) (2025.8.3)
Requirement already satisfied: markdown>=2.6.8 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorboard~=2.20.0->tensorflow<2.21,>=2.20->tf-keras) (3.8.2)
Requirement already satisfied: pillow in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorboard~=2.20.0->tensorflow<2.21,>=2.20->tf-keras) (11.3.0)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorboard~=2.20.0->tensorflow<2.21,>=2.20->tf-keras) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from tensorboard~=2.20.0->tensorflow<2.21,>=2.20->tf-keras) (3.1.3)
Requirement already satisfied: rich in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from keras>=3.10.0->tensorflow<2.21,>=2.20->tf-keras) (14.1.0)
Requirement already satisfied: namex in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from keras>=3.10.0->tensorflow<2.21,>=2.20->tf-keras) (0.1.0)
Requirement already satisfied: optree in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from keras>=3.10.0->tensorflow<2.21,>=2.20->tf-keras) (0.19.1)
Requirement already satisfied: importlib-metadata>=4.4 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from markdown>=2.6.8->tensorboard~=2.20.0->tensorflow<2.21,>=2.20->tf-keras) (8.7.0)
Requirement already satisfied: zipp>=3.20 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from importlib-metadata>=4.4->markdown>=2.6.8->tensorboard~=2.20.0->tensorflow<2.21,>=2.20->tf-keras) (3.23.0)
Requirement already satisfied: MarkupSafe>=2.1.1 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from werkzeug>=1.0.1->tensorboard~=2.20.0->tensorflow<2.21,>=2.20->tf-keras) (3.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from rich->keras>=3.10.0->tensorflow<2.21,>=2.20->tf-keras) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from rich->keras>=3.10.0->tensorflow<2.21,>=2.20->tf-keras) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow<2.21,>=2.20->tf-keras) (0.1.2)
```

In [3]:
```
!pip install nltk
```

```
Requirement already satisfied: nltk in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (3.9.2)
Requirement already satisfied: click in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from nltk) (8.1.8)
Requirement already satisfied: joblib in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from nltk) (1.5.1)
Requirement already satisfied: regex>=2021.8.3 in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from nltk) (2026.1.15)
Requirement already satisfied: tqdm in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from nltk) (4.67.3)
Requirement already satisfied: colorama in c:\users\grecl\anaconda3\envs\tf_env\lib\site-packages (from click->nltk) (0.4.6)
```

In [4]:
```
#import libraries
import pandas as pd # load, clean, manipulate tables
import numpy as np # fast numerical operations
import matplotlib.pyplot as plt # Basic plotting (bar charts, histograms, etc.)
import seaborn as sns # More advanced, prettier statistical plots.
from sklearn.preprocessing import LabelEncoder
from sklearn.feature_extraction.text import CountVectorizer # Converts text → bag-of-
from sklearn.feature_extraction.text import TfidfVectorizer # Converts text → TF-IDF
from sklearn.model_selection import train_test_split, RandomizedSearchCV, GridSearchC
                    # GridSearchCV - Tries every possible combination of hyperpara
                    # Randomized Search CV - Picks random combinations of hyperpa
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC # ML Algorithm to find the best possible boundary to sepa
from sklearn.utils import class_weight # Balance classes
from sklearn.naive_bayes import MultinomialNB # Multinomial Naive Bayes - learn word
from sklearn.neighbors import KNeighborsClassifier # Predicts a label by looking at t
from sklearn.tree import DecisionTreeClassifier # learns a series of if-else rules th
from imblearn.over_sampling import SMOTE # fix imbalanced datasets by creating synthe
from imblearn.pipeline import make_pipeline, Pipeline # chain multiple steps (preproc
import plotly.graph_objects as go # build highly customized, interactive visualizatio
import plotly.express as px # Plotly's high-level, fast, simple plotting interface.
from scipy.stats import randint
from tensorflow.keras.preprocessing.text import Tokenizer # converts raw text into sm
from tensorflow.keras.preprocessing.sequence import pad_sequences # adds zeros (or an
from tensorflow.keras.models import Sequential # Sequential is a model where each lay
from tensorflow.keras.layers import Embedding, LSTM, Bidirectional, Dense, Dropout
                    # Embedding - Converts each integer token into a dense vector t
                    # LSTM - (LongShortTermMemory)Learns patterns in sequences — esp
                    # Bidirectional - Runs LSTM forward and backward over the sente
                    # Dense layer - A fully connected layer that makes final predic
                    # Dropout - Randomly turns off some neurons during training to p
from tensorflow.keras.optimizers import Adam, AdamW # used to update neural network w
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
                    # EarlyStopping - Stops training automatically when the model s
                    # ReduceLROnPlateau - automatically reduces the learning rate w
from tensorflow.keras.layers import Input # defines the entry point of a neural netwo
from tensorflow.keras.utils import to_categorical # converts integer class labels int
from keras_tuner import RandomSearch
from transformers import DistilBertTokenizerFast
from transformers import DistilBertForSequenceClassification
from transformers import TrainingArguments
from transformers import Trainer

from sklearn.dummy import DummyClassifier # a baseline classifier

import re # Regular expressions for text cleaning (remove URLs, symbols, etc.)
import datasets

import nltk # natural language toolkit
from nltk.corpus import stopwords # Common words
from nltk.tokenize import word_tokenize # splitting text into words or subunits (toke
from nltk.stem import PorterStemmer # reducing words to their root form
from nltk.stem import WordNetLemmatizer # reduces words to dictionary/base form
# from nltk.sentiment.vader import SentimentIntensityAnalyzer

nltk.download('stopwords')
nltk.download('punkt') # pre-trained tokenizer model in NLTK
nltk.download('wordnet') # WordNet is a large lexical database of English words.
# nltk.download('vader_lexicon')
nltk.download('punkt_tab') # tokenization of tabular text

import warnings
warnings.filterwarnings('ignore')

%matplotlib inline
```

```
c:\Users\grecl\anaconda3\envs\tf_env\lib\site-packages\tqdm\auto.py:21: TqdmWarning: I
Progress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
WARNING:tensorflow:From c:\Users\grecl\anaconda3\envs\tf_env\lib\site-packages\tf_keras\src\losses.py:2976: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses.sparse_softmax_cross_entropy instead.
```

```
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\grecl\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\grecl\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\grecl\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\grecl\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

1.a) **Loading and Inspecting the Dataset**

In [5]:
```
#Loading the Dataset
data = pd.read_csv('judge-1377884607_tweet_product_company.csv', encoding='latin1')
data.head()
```

Out[5]:

	tweet_text	emotion_in_tweet_is_directed_at	is_there_an_emotion_directed_at_a_brand_or_product
0	.@wesley83 I have a 3G iPhone. After 3 hrs twe...	iPhone	Negative emotion
1	@jessedee Know about @fludapp ? Awesome iPad/i...	iPad or iPhone App	Positive emotion
2	@swonderlin Can not wait for #iPad 2 also. The...	iPad	Positive emotion
3	@sxsw I hope this year's festival isn't as cra...	iPad or iPhone App	Negative emotion
4	@sxtxstate great stuff on Fri #SXSW: Marissa M...	Google	Positive emotion

In [6]:
```
# Checking the data structure
data.shape
```

Out[6]:
```
(9093, 3)
```

In [7]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9093 entries, 0 to 9092
Data columns (total 3 columns):
 #   Column                                              Non-Null Count  Dtype
---  ------                                              --------------  -----
 0   tweet_text                                          9092 non-null   object
 1   emotion_in_tweet_is_directed_at                     3291 non-null   object
 2   is_there_an_emotion_directed_at_a_brand_or_product  9093 non-null   object
dtypes: object(3)
memory usage: 213.2+ KB
```

In [8]:
```
#Renaming the column names for consistency
data.rename(columns={
    'tweet_text':'tweet',
    'emotion_in_tweet_is_directed_at':'emotion',
    'is_there_an_emotion_directed_at_a_brand_or_product':'label'
}, inplace =True)
```

In [9]:
```
#Confirming the renamed columns
data.columns
```

Out[9]:
```
Index(['tweet', 'emotion', 'label'], dtype='object')
```

b).**Exploring and Understanding the Data**

In [10]:
```
#checking class imbalance
data['label'].value_counts()
```

Out[10]:
```
label
No emotion toward brand or product    5389
Positive emotion                      2978
Negative emotion                       570
I can't tell                           156
Name: count, dtype: int64
```

In [11]:
```
#Checking for duplicates and dropping the duplicates
data.duplicated().sum()
```

Out[11]:
```
22
```

In [12]:
```
data=data.drop_duplicates()
```

In [13]:
```
data.shape
```

Out[13]:
```
(9071, 3)
```

In [14]:
```
data.isnull().sum()
```

Out[14]:
```
tweet         1
emotion    5789
label         0
dtype: int64
```

Handling the missing values

In [15]:
```
# Dropping the row with missing value in tweet.
df=data.dropna(subset=['tweet'])
```

In [16]:
```
#For 'emotion' has many missing values(5789) more than half.
#Filling with a placeholder
df['emotion'] = data['emotion'].fillna('unspecified')
```

In [17]:
```
#Confirming missing values
df.isnull().sum()
```

Out[17]:
```
tweet      0
emotion    0
label      0
dtype: int64
```

In [18]:
```
#Exploring the unique and target variable
print("Unique labels:", df['label'].unique()) # Unique
```

```
Unique labels: ['Negative emotion' 'Positive emotion'
 'No emotion toward brand or product' "I can't tell"]
```

In [19]:
```
#Exploring the target variable
print('sentiment count:')
sentiment_count =df['label'].value_counts()
sentiment_count
```

```
sentiment count:
```

Out[19]:
```
label
No emotion toward brand or product    5375
Positive emotion                      2970
Negative emotion                       569
I can't tell                           156
Name: count, dtype: int64
```

In [20]:
```
#Sentiment Distribution
sentiment_percent = df['label'].value_counts(normalize=True)
sentiment_percent
```

Out[20]:
```
label
No emotion toward brand or product    0.592613
Positive emotion                      0.327453
Negative emotion                      0.062734
I can't tell                          0.017200
Name: proportion, dtype: float64
```

In [21]:
```
#Brand Distribution
print("\nTop Brands Mentioned:")
brand_counts =df['emotion'].value_counts().head(10)
print(brand_counts)
```

```
Top Brands Mentioned:
emotion
unspecified                        5788
iPad                                945
Apple                               659
iPad or iPhone App                  469
Google                              428
iPhone                              296
Other Google product or service     293
Android App                          80
Android                              77
Other Apple product or service       35
Name: count, dtype: int64
```

2.a) **Exploratory Data Analysis**

In [22]:
```
# Sentiment Distribution
plt.figure(figsize=(10,8))
sns.countplot(x = "label",data =df,order=sentiment_percent.index,hue='label',palette=
plt.title('Distribution of Emotions Directed at a Brand or Product',fontsize=14,fontwe
plt.xlabel('Emotion Category',fontsize=14,fontweight='bold')
plt.ylabel('Count',fontsize =14,fontweight='bold')
plt.xticks(rotation =45,ha='right',fontweight ='bold')
plt.tight_layout()
plt.show()
```

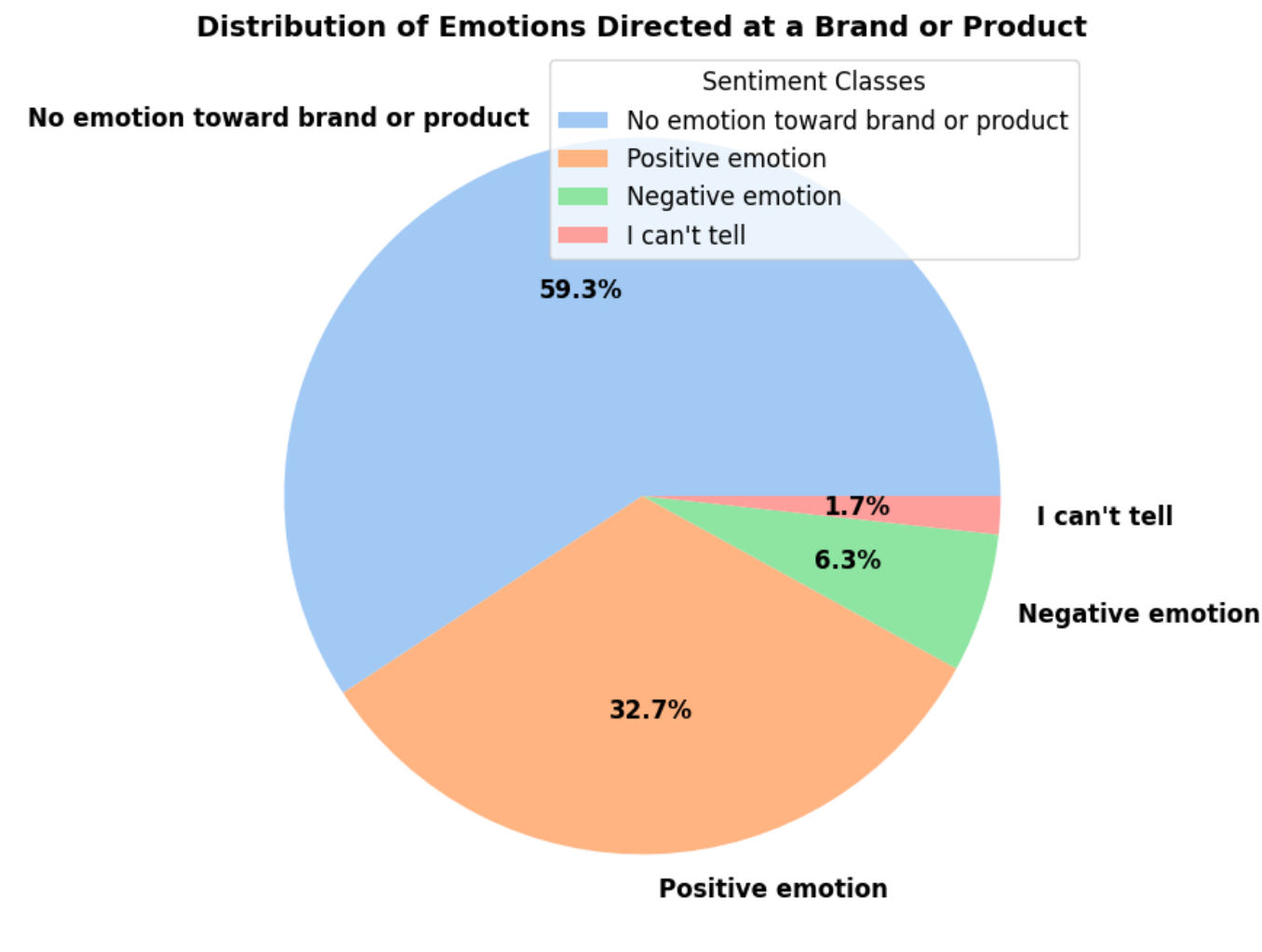
In [22]:
```
# Sentiment distribution percentage
%matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns

# Sentiment Distribution (%)
plt.figure(figsize=(10,8))
df["label"].value_counts().plot.pie(
    autopct="%1.1f%%",
    colors=sns.color_palette("pastel"),
    textprops={'fontsize':12, 'fontweight':'bold'}
)

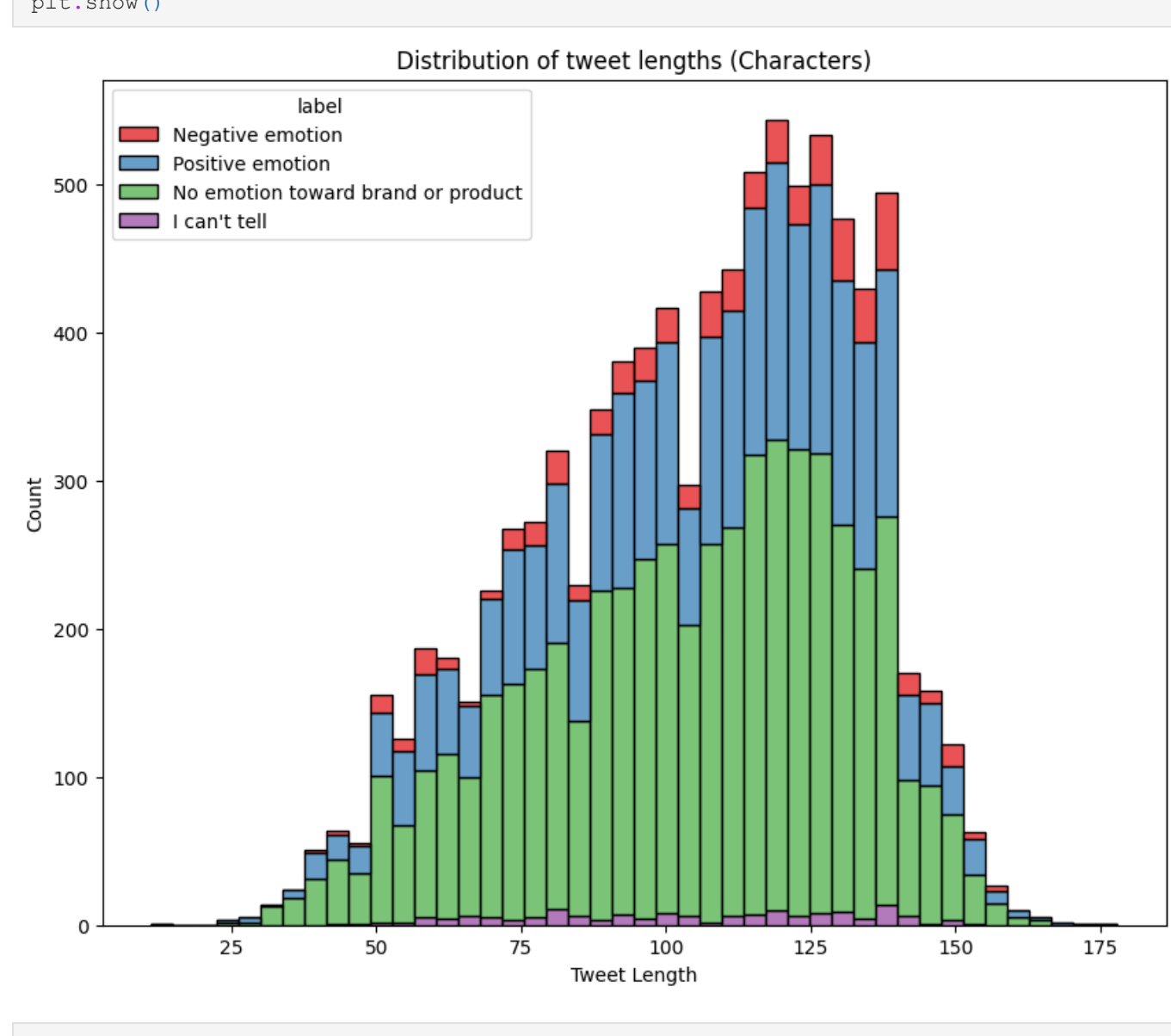
plt.title("Distribution of Emotions Directed at a Brand or Product",
          fontsize=14, fontweight='bold')
plt.ylabel("")  # remove y-axis label

# Add legend with class names
plt.legend(labels=df["label"].value_counts().index,
           title="Sentiment Classes",
           loc="upper right",
           fontsize=12,
           title_fontsize=12)

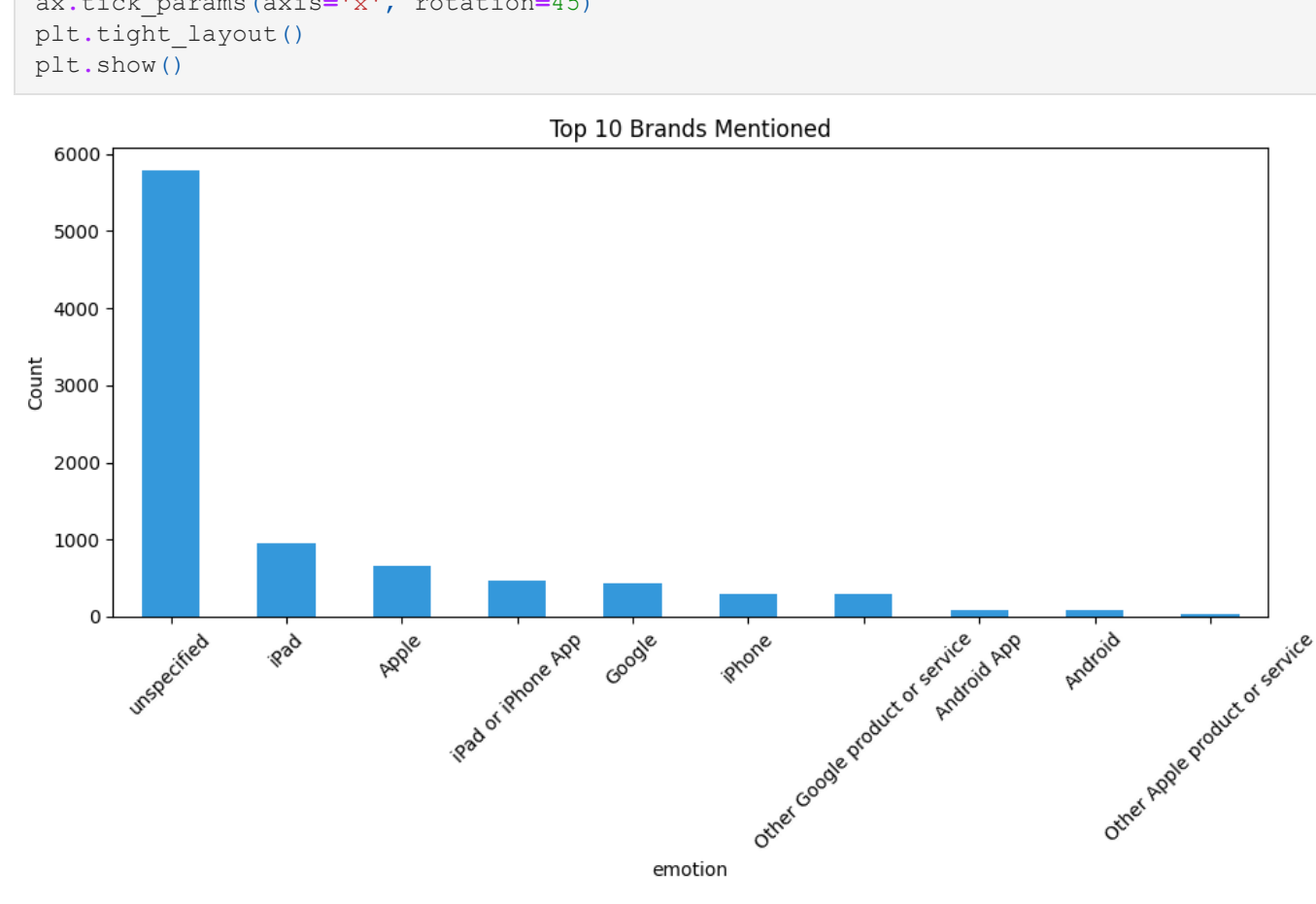
plt.show()
```



In [23]:
```python
# Distribution of tweet length
df['tweet_length'] = df['tweet'].astype(str).apply(len)

plt.figure(figsize=(10,8))
sns.histplot(
    data=df,
    x="tweet_length",
    hue="label",
    palette='Set1',  # <- change palette to a categorical one
    multiple="stack"  # optional: stacks bars for each category
)
plt.title("Distribution of tweet lengths (Characters)")
plt.xlabel("Tweet Length")
plt.ylabel("Count")
plt.show()
```

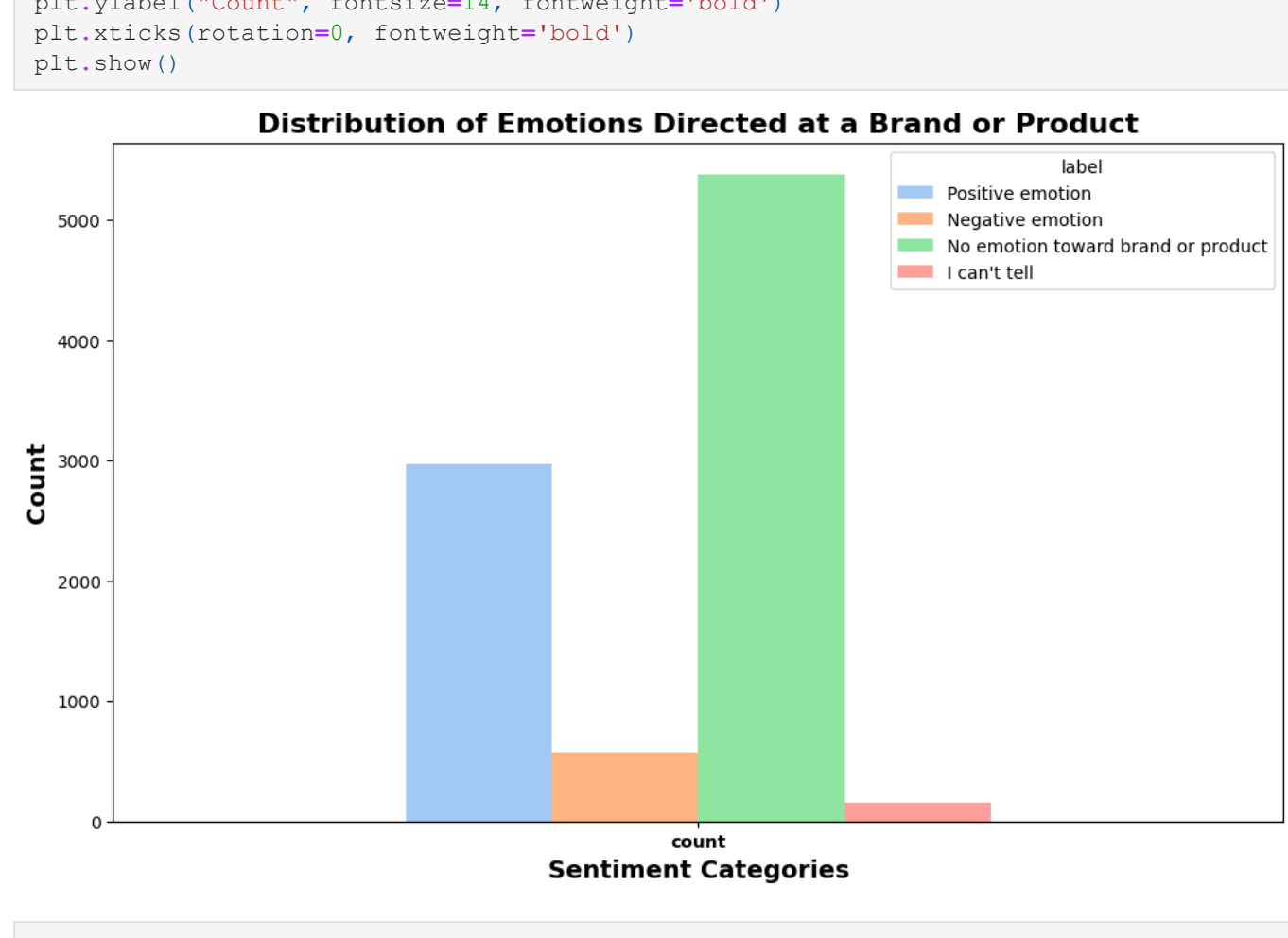


In [24]:
```python
# Brands Distribution
top_brands = df['emotion'].value_counts()

fig, ax = plt.subplots(figsize=(10, 6))
top_brands.plot(kind='bar', ax=ax, color='#3498db')
ax.set_title('Top 10 Brands Mentioned')
ax.set_ylabel('Count')
ax.tick_params(axis='x', rotation=45)
plt.tight_layout()
plt.show()
```



In [25]:
```python
# Count sentiment categories
brand_sentiment = df['label'].value_counts().to_frame().T
all_sentiments = [
    'Positive emotion',
    'Negative emotion',
    'No emotion toward brand or product',
    "I can't tell"
]
brand_sentiment = brand_sentiment.reindex(columns=all_sentiments, fill_value=0)
```

In [26]:
```python
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(12, 7))
brand_sentiment.plot(kind='bar', ax=ax, color=sns.color_palette("pastel"))
plt.title("Distribution of Emotions Directed at a Brand or Product",
          fontsize=16, fontweight='bold')
plt.xlabel("Sentiment Categories", fontsize=14, fontweight='bold')
plt.ylabel("Count", fontsize=14, fontweight='bold')
plt.xticks(rotation=0, fontweight='bold')
plt.show()
```



In [27]:
```python
# Compare sentiment across different brands
brand_sentiment_pivot = df.pivot_table(
    index='emotion',
    columns='label',
    values='tweet',
    aggfunc='count',
    fill_value=0
)

# Calculate net sentiment score per brand
brand_sentiment_pivot['net_sentiment'] = (
    brand_sentiment_pivot['Positive emotion'] -
    brand_sentiment_pivot['Negative emotion']
)
brand_sentiment_pivot
```

Out[27]:

label	I can't tell	Negative emotion	No emotion toward brand or product	Positive emotion	net_sentiment
emotion					
Android	0	8	1	68	60
Android App	0	8	1	71	63
Apple	2	95	21	541	446
Google	1	68	15	344	276
Other Apple product or service	0	2	1	32	30
Other Google product or service	1	47	9	236	189
iPad	4	125	24	792	667
iPad or iPhone App	0	63	10	396	333
iPhone	1	102	9	184	82
unspecified	147	51	5284	306	255

3.a) **Data Preparation**

In [28]:
```python
# Data Cleaning and Preprocessing
def clean_text(text):
    if not isinstance(text, str):
        return "" # Handle non-string inputs gracefully, e.g., for NaN values in the 
    # Convert to lowercase
    text = text.lower()
    # Remove URLs
    text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)
    # Remove symbols
    text = re.sub(r'@\w+|#\w+', '', text)
    # Remove punctuation and numbers
    text = re.sub(r'[^\w\s]', ' ', text)
    text = re.sub(r'\d+', '', text)

    # Remove extra whitespace
    text = ' '.join(text.split())

    # Tokenize
    tokens = word_tokenize(text)

    # Remove stopwords
    stop_words = set(stopwords.words('english'))
    tokens = [word for word in tokens if word not in stop_words]

    # Initialize stemmer and lemmatizer
    stemmer = PorterStemmer()
    lemmatizer = WordNetLemmatizer()

    # Apply stemming
    stemmed_tokens = [stemmer.stem(word) for word in tokens]

    # Apply lemmatization to stemmed tokens
    final_tokens = [lemmatizer.lemmatize(word) for word in stemmed_tokens]

    # Join tokens back to text
    cleaned_text = ' '.join(final_tokens)

    return cleaned_text

print("Text cleaning completed!")
print(f"Remaining samples: {len(df)}")
```

```
Text cleaning completed!
Remaining samples: 9070
```

In [29]:
```python
#Apply cleaning to dataset
df.loc[:,'cleaned_text'] = df['tweet'].apply(clean_text)

# Remove empty texts after cleaning
df = df[df['cleaned_text'].str.len() > 0]


df[['tweet', 'cleaned_text']].head()
```

Out[29]:

	tweet	cleaned_text
0	.@wesley83 I have a 3G iPhone. After 3 hrs twe...	g iphon hr tweet dead need upgrad plugin station
1	@jessedee Know about @fludapp ? Awesome iPad/i...	know awesom ipad iphon app like appreci design...
2	@swonderlin Can not wait for #iPad 2 also. The...	wait also sale
3	@sxsw I hope this year's festival isn't as cra...	hope year festiv crashi year iphon app
4	@sxtxstate great stuff on Fri #SXSW: Marissa M...	great stuff fri marissa mayer googl tim reilli...

3.b) **Feature Engineering**

In [30]:
```python
#Prepare target variable
#Combine 'I can't tell' with 'No emotion' for cleaner classification
df['sentiment'] = df['label'].replace({'I can\'t tell': 'No emotion toward brand or p

# Filter to only include the three main sentiment categories
main_sentiments = ['Positive emotion', 'Negative emotion', 'No emotion toward brand o
df = df[df['sentiment'].isin(main_sentiments)]

# Encode target variable
label_encoder = LabelEncoder()
df['emotion'] = label_encoder.fit_transform(df['sentiment'])

print("Target variable prepared!")
print("Sentiment mapping:", dict(zip(label_encoder.classes_, label_encoder.transform(
print("\nFinal distribution:")
print(df['sentiment'].value_counts())
```

```
Target variable prepared!
Sentiment mapping: {'Negative emotion': 0, 'No emotion toward brand or product': 1, 'P
ositive emotion': 2}

Final distribution:
sentiment
No emotion toward brand or product    5529
Positive emotion                      2970
Negative emotion                       569
Name: count, dtype: int64
```

In [31]:
```python
df.shape
```

Out[31]:
```
(9068, 6)
```

In [32]:
```python
df.columns
```

Out[32]:
```
Index(['tweet', 'emotion', 'label', 'tweet_length', 'cleaned_text',
       'sentiment'],
      dtype='object')
```

In [33]:
```python
# Define brand mapping dictionary
brand_mapping = {
    # Apple group
    'Apple': 'Apple',
    'iPad': 'Apple',
    'iPhone': 'Apple',
    'imac': 'Apple',
    'iPad or iPhone App': 'Apple',
    'Other Apple product or service': 'Apple',

    # Google group
    'Google': 'Google',
    'Android': 'Google',
    'Android App': 'Google',
    'Other Google product or service': 'Google',

    # Others
    'Not Specified': 'Not Specified'
}

# Apply the mapping
df['brand_group'] = df['emotion'].map(brand_mapping)

# Fill any NaN (for brands not in mapping) with original value
df['brand_group'] = df['brand_group'].fillna(df['emotion'])
df['brand_group'].value_counts()
```

Out[33]:
```
brand_group
1    5529
2    2970
0     569
Name: count, dtype: int64
```

In [34]:
```python
df.head()
```

Out[34]:

	tweet	emotion	label	tweet_length	cleaned_text	sentiment	brand_group
0	.@wesley83 I have a 3G iPhone. After 3 hrs twe...	0	Negative emotion	127	g iphon hr tweet dead need upgrad plugin station	Negative emotion	0
1	@jessedee Know about @fludapp ? Awesome iPad/i...	2	Positive emotion	139	know awesom ipad iphon app like appreci design...	Positive emotion	2
2	@swonderlin Can not wait for #iPad 2 also. The...	2	Positive emotion	79	wait also sale	Positive emotion	2
3	@sxsw I hope this year's festival isn't as cra...	0	Negative emotion	82	hope year festiv crashi year iphon app	Negative emotion	0
4	@sxtxstate great stuff on Fri #SXSW: Marissa M...	2	Positive emotion	131	great stuff fri marissa mayer googl tim reilli...	Positive emotion	2

In [35]:
```python
df = df.drop(columns=['label','emotion'],errors='ignore')   # redundant raw labels
```

In [36]:
```python
df.columns
```

Out[36]:
```
Index(['tweet', 'tweet_length', 'cleaned_text', 'sentiment', 'brand_group'], dtype='ob
ject')
```

In [37]:
```python
df.head()
```

Out[37]:

	tweet	tweet_length	cleaned_text	sentiment	brand_group
0	.@wesley83 I have a 3G iPhone. After 3 hrs twe...	127	g iphon hr tweet dead need upgrad plugin station	Negative emotion	0
1	@jessedee Know about @fludapp ? Awesome iPad/i...	139	know awesom ipad iphon app like appreci design...	Positive emotion	2
2	@swonderlin Can not wait for #iPad 2 also. The...	79	wait also sale	Positive emotion	2
3	@sxsw I hope this year's festival isn't as cra...	82	hope year festiv crashi year iphon app	Negative emotion	0
4	@sxtxstate great stuff on Fri #SXSW: Marissa M...	131	great stuff fri marissa mayer googl tim reilli...	Positive emotion	2

In [38]:
```python
# Encoding the sentiment to numerical
from sklearn.preprocessing import LabelEncoder

label_encoder = LabelEncoder()
df['sentiment_emotion'] = label_encoder.fit_transform(df['sentiment'])
```

In [39]:
```python
df.head()
```

Out[39]:

	tweet	tweet_length	cleaned_text	sentiment	brand_group	sentiment_emotion
0	.@wesley83 I have a 3G iPhone. After 3 hrs twe...	127	g iphon hr tweet dead need upgrad plugin station	Negative emotion	0	0
1	@jessedee Know about @fludapp ? Awesome iPad/i...	139	know awesom ipad iphon app like appreci design...	Positive emotion	2	2
2	@swonderlin Can not wait for #iPad 2 also. The...	79	wait also sale	Positive emotion	2	2
3	@sxsw I hope this year's festival isn't as cra...	82	hope year festiv crashi year iphon app	Negative emotion	0	0
4	@sxtxstate great stuff on Fri #SXSW: Marissa M...	131	great stuff fri marissa mayer googl tim reilli...	Positive emotion	2	2

1. a). **Preparing Data for Modeling**

In [40]:
```python
# Preparing the data for modeling
X = df['cleaned_text']
y = df['sentiment_emotion']

# Split the data
X_train,X_test,y_train,y_test =train_test_split(X,y,test_size=0.2,random_state=42)
print(f"Training set: {X_train.shape[0]} samples")
print(f"Test set:{X_test.shape[0]} samples")
```

```
Training set: 7254 samples
Test set:1814 samples
```

In [41]:
```python
# TF-IDF Vectorization with n-grams
#fiting TF-IDF on training text and transforming
tfidf =TfidfVectorizer()
X_train_vec= tfidf.fit_transform(X_train)
X_test_vec= tfidf.transform(X_test)

print("Vectorization Complete")
```

```
Vectorization Complete
```

In [42]:
```python
print(X_train_vec)
```

```
  (0, 5586)	0.26316857881136896
  (0, 3281)	0.1679570029825773
  (0, 5379)	0.44230043549323
  (0, 218)	0.34717027541223283
  (0, 2856)	0.28971387064956966
  (0, 2551)	0.15646621046422649
  (0, 4661)	0.15102592570177173
  (0, 183)	0.21355408002516135
  (0, 2998)	0.4325612055906897
  (0, 2021)	0.20889382175731572
  (0, 2342)	0.35221658910097303
  (0, 4943)	0.23397337610628985
  (1, 4661)	0.14730728852663158
  (1, 2549)	0.2414911062459586
  (1, 41)	0.42191046352664613
  (1, 2841)	0.2105007949034364
  (1, 3426)	0.20005898352474694
  (1, 5417)	0.43140988915581724
  (1, 3108)	0.35918036010193705
  (1, 360)	0.33805457713130244
  (1, 5317)	0.2869508310480568
  (1, 2057)	0.21311801882716513
  (1, 224)	0.13603034373633127
  (1, 4833)	0.28630358556086216
  (2, 4661)	0.1510977705516458
  :	:
  (7250, 5137)	0.48528195096212845
  (7251, 3281)	0.11430213950287063
  (7251, 4661)	0.10277979555222706
  (7251, 2021)	0.14216144805983072
  (7251, 224)	0.09491160320686817
  (7251, 4152)	0.08135306658085686
  (7251, 5299)	0.17986857962974417
  (7251, 2959)	0.17134348466851657
  (7251, 204)	0.1959306212382942
  (7251, 49)	0.23240387980817642
  (7251, 3690)	0.25443886577777636
  (7251, 1387)	0.2887730829235874
  (7251, 3880)	0.2796369951549569
  (7251, 2370)	0.6020098618355547
  (7251, 2304)	0.31957508530610823
  (7251, 1070)	0.3091169167028984
  (7252, 2843)	0.14869490925261408
  (7252, 2907)	0.40980727150722046
  (7252, 930)	0.47606655950796184
  (7252, 5208)	0.5165095435091488
  (7252, 5001)	0.562606742373736
  (7253, 2549)	0.18054400957006372
  (7253, 4409)	0.529636875956958
  (7253, 27)	0.49804868521886386
  (7253, 1328)	0.6624470902545198
```

b).**Iteration 0 : (Baseline Model)**

In [43]:
```python
# Baseline Model - Most Frequent Class
print("BASELINE MODEL")

baseline = DummyClassifier(strategy= 'most_frequent')
baseline.fit(X_train,y_train)

# Predictions
y_pred_baseline = baseline.predict(X_test)

# Evaluation
baseline_f1 =f1_score(y_test, y_pred_baseline, average='macro')
baseline_accuracy = accuracy_score(y_test,y_pred_baseline)

print(f"Baseline Model (Most Frequent Class)")
print(f"Macro F1-score: {baseline_f1: .4f}")
print(f"Accuracy: {baseline_accuracy:.4f}")
print(f"\nClassification Report:")
print(classification_report(y_test,y_pred_baseline,target_names= label_encoder.classe
```

```
BASELINE MODEL
Baseline Model (Most Frequent Class)
Macro F1-score:  0.2542
Accuracy: 0.6163

Classification Report:
                                    precision    recall  f1-score   support

                  Negative emotion       0.00      0.00      0.00       124
No emotion toward brand or product       0.62      1.00      0.76      1118
                  Positive emotion       0.00      0.00      0.00       572

                          accuracy                           0.62      1814
                         macro avg       0.21      0.33      0.25      1814
                      weighted avg       0.38      0.62      0.47      1814
```

c).**Iteration 1: Logistic Regression Model**

In [44]:
```python
# Logistic Regression as Baseline Model
lr_model = LogisticRegression(class_weight='balanced',max_iter=1000)
lr_model.fit(X_train_vec,y_train)
```

Out[44]:
```
LogisticRegression
LogisticRegression(class_weight='balanced', max_iter=1000)
```

In [45]:
```python
# Predictions on the training data
y_train_lr_pred = lr_model.predict(X_train_vec)

# Evaluate based on
print(f"Accuracy: {accuracy_score(y_train,y_train_lr_pred):.2f}")
print("Confusion Matrix:")
print(confusion_matrix(y_train,y_train_lr_pred))
print("\nClassification Report:")
print(classification_report(y_train,y_train_lr_pred))
```

```
Accuracy: 0.79
Confusion Matrix:
[[ 439    4    2]
 [ 303 3410  698]
 [  95  403 1900]]

Classification Report:
              precision    recall  f1-score   support

           0       0.52      0.99      0.68       445
           1       0.89      0.77      0.83      4411
           2       0.73      0.79      0.76      2398

    accuracy                           0.79      7254
   macro avg       0.72      0.85      0.76      7254
weighted avg       0.82      0.79      0.80      7254
```

In [46]:
```python
#Predictions on the test data
y_test_lr_pred = lr_model.predict(X_test_vec)

#Evaluation
lr_f1 = f1_score(y_test,y_test_lr_pred,average='macro')
lr_accuracy =accuracy_score(y_test,y_test_lr_pred)


print(f"Accuracy: {accuracy_score(y_test,y_test_lr_pred):.2f}")
print("Confusion Matrix:")
print(confusion_matrix(y_test,y_test_lr_pred))
print("\nClassification Report:")
print(classification_report(y_test,y_test_lr_pred))
```

```
Accuracy: 0.66
Confusion Matrix:
[[ 68  34  22]
 [ 92 769 257]
 [ 34 181 357]]

Classification Report:
              precision    recall  f1-score   support

           0       0.35      0.55      0.43       124
           1       0.78      0.69      0.73      1118
           2       0.56      0.62      0.59       572

    accuracy                           0.66      1814
   macro avg       0.56      0.62      0.58      1814
weighted avg       0.68      0.66      0.67      1814
```

d). **Logistic Regression Model Tuning**

In [47]:
```python
# Model Tuning
#Define model
lr = LogisticRegression(max_iter=2000)
# Parameter grid
param_grid ={
    'C': [0.001,0.01,0.1,1,10],
    'penalty':['l1','l2'],
    'solver':['liblinear','saga'],
    'class_weight':['balanced',None]
}

# Grid search
grid = GridSearchCV(
    lr,
    param_grid,
    scoring= 'f1_macro',
    cv=5,
    n_jobs=-1,
    verbose=2
)

grid.fit(X_train_vec,y_train)

print("Best Params:", grid.best_params_)
print("Best Score:", grid.best_score_)
```

```
Fitting 5 folds for each of 40 candidates, totalling 200 fits
Best Params: {'C': 10, 'class_weight': 'balanced', 'penalty': 'l2', 'solver': 'libline
ar'}
Best Score: 0.5701932549623802
```

In [48]:
```python
# Fitting the model
lr.set_params(**grid.best_params_)
lr.fit(X_train_vec,y_train)

# Predictions for test data
y_test_lr_pred_tuned = lr.predict(X_test_vec)

# Evaluation
lr_f1_tuned =f1_score(y_test,y_test_lr_pred_tuned,average='macro')
lr_accuracy_tuned = accuracy_score(y_test, y_test_lr_pred_tuned)

print(f"Accuracy: {accuracy_score(y_test, y_test_lr_pred_tuned):.2f}")
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_test_lr_pred_tuned))
print("\nClassification Report:")
print(classification_report(y_test, y_test_lr_pred_tuned))
```

```
Accuracy: 0.69
Confusion Matrix:
[[ 49  50  25]
 [ 47 873 198]
 [ 14 220 338]]

Classification Report:
              precision    recall  f1-score   support

           0       0.45      0.40      0.42       124
           1       0.76      0.78      0.77      1118
           2       0.60      0.59      0.60       572

    accuracy                           0.69      1814
   macro avg       0.60      0.59      0.60      1814
weighted avg       0.69      0.69      0.69      1814
```

e).**Hyperparameter Tuning**

In [49]:
```python
tfidf = TfidfVectorizer(
        lowercase=True,
        max_features=5000, # limit the number of features to avoid overfitting
        ngram_range=(1, 2),  # Include unigrams and bigrams
        stop_words='english',
        min_df=3,
        max_df=0.9

)
# Transform text data
X_train_tfidf = tfidf.fit_transform(X_train)
X_test_tfidf = tfidf.transform(X_test)

print(f"TF-IDF Features: {X_train_tfidf.shape[1]}")
```

```
TF-IDF Features: 5000
```

f). **Iteration 2 : Random Forest Classifier**

In [50]:
```python
#Instantiating the Random Forest Classifier
rf_model = RandomForestClassifier(
    n_estimators=100,          # number of trees
    max_depth=None,            # allow full tree growth
    class_weight='balanced',   # paying attention to the imbalance
    random_state=42,
    n_jobs=-1                  # use all CPU cores
)

# Fitting the Classifier
rf_model.fit(X_train_tfidf, y_train)
```

Out[50]:
```
RandomForestClassifier
RandomForestClassifier(class_weight='balanced', n_jobs=-1, random_state=42)
```

In [51]:
```python
#Predictions
y_pred_rf =rf_model.predict(X_test_tfidf)

# Evaluation
rf_f1 = f1_score(y_test,y_pred_rf,average='macro')
rf_accuracy = accuracy_score(y_test,y_pred_rf)

print("random Forest with TF-IDf")
print(f"Macro F1-Score: {rf_f1:.4f}")
print(f"Accuracy: {rf_accuracy:.4f}")
print(f"Improvement over baseline: {rf_f1 - baseline_f1:.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred_rf, target_names=label_encoder.classes_))
```

```
random Forest with TF-IDf
Macro F1-Score: 0.5334
Accuracy: 0.6808
Improvement over baseline: 0.2792

Classification Report:
                                    precision    recall  f1-score   support

                  Negative emotion       0.55      0.22      0.31       124
No emotion toward brand or product       0.71      0.85      0.77      1118
                  Positive emotion       0.60      0.45      0.51       572

                          accuracy                           0.68      1814
                         macro avg       0.62      0.51      0.53      1814
                      weighted avg       0.67      0.68      0.66      1814
```

Tuning Random Forest Model

In [52]:
```python
# rf helper functions

def evaluate_model(model, X_test, y_test, label_encoder, baseline_f1=None, title="Mode
    y_pred = model.predict(X_test)

    f1 = f1_score(y_test, y_pred, average='macro')
    acc = accuracy_score(y_test, y_pred)

    print("\n" + title)
    print(f"Macro F1-Score: {f1:.4f}")
    print(f"Accuracy: {acc:.4f}")

    if baseline_f1 is not None:
        print(f"Improvement over baseline: {f1 - baseline_f1:.4f}")

    print("\nClassification Report:")
    print(classification_report(y_test, y_pred, target_names=label_encoder.classes_))

    return f1, acc


def plot_top_features(best_rf, vectorizer, top_n=30):
    importances = best_rf.named_steps['rf'].feature_importances_
    indices = np.argsort(importances)[::-1]

    top_features = np.array(vectorizer.get_feature_names_out())[indices][:top_n]
    top_scores = importances[indices][:top_n]

    return pd.DataFrame({
        "Feature": top_features,
        "Importance": top_scores
    })


# RandomisedSearchCV Tuning
# Using the tfidf vectorizer already fitted
pipeline = Pipeline([
    ('rf', RandomForestClassifier(random_state=42, class_weight='balanced'))
])

param_dist = {
    'rf__n_estimators': randint(100, 500),   # Number of trees in the forest
    'rf__max_depth': randint(10, 50),        # Maximum depth of the tree
    'rf__min_samples_split': randint(2, 20),  # Minimum number of samples required to
    'rf__min_samples_leaf': randint(1, 10),   # Minimum number of samples required to
    'rf__bootstrap': [True, False]            # Whether bootstrap samples are used whe
}

# Initialize RandomizedSearchCV
rf_search = RandomizedSearchCV(
    estimator=pipeline,
    param_distributions=param_dist,
    n_iter=50,  # Number of parameter settings that are sampled
    cv=3,       # Number of folds in cross-validation
    verbose=2,   # Controls the verbosity: higher values means more messages
    random_state=42,
    n_jobs=-1   # Use all available cores
)


# Fit RandomizedSearchCV
rf_search.fit(X_train_tfidf, y_train)

print("\nBest parameters:", rf_search.best_params_)
print("Best CV score:", rf_search.best_score_)
```

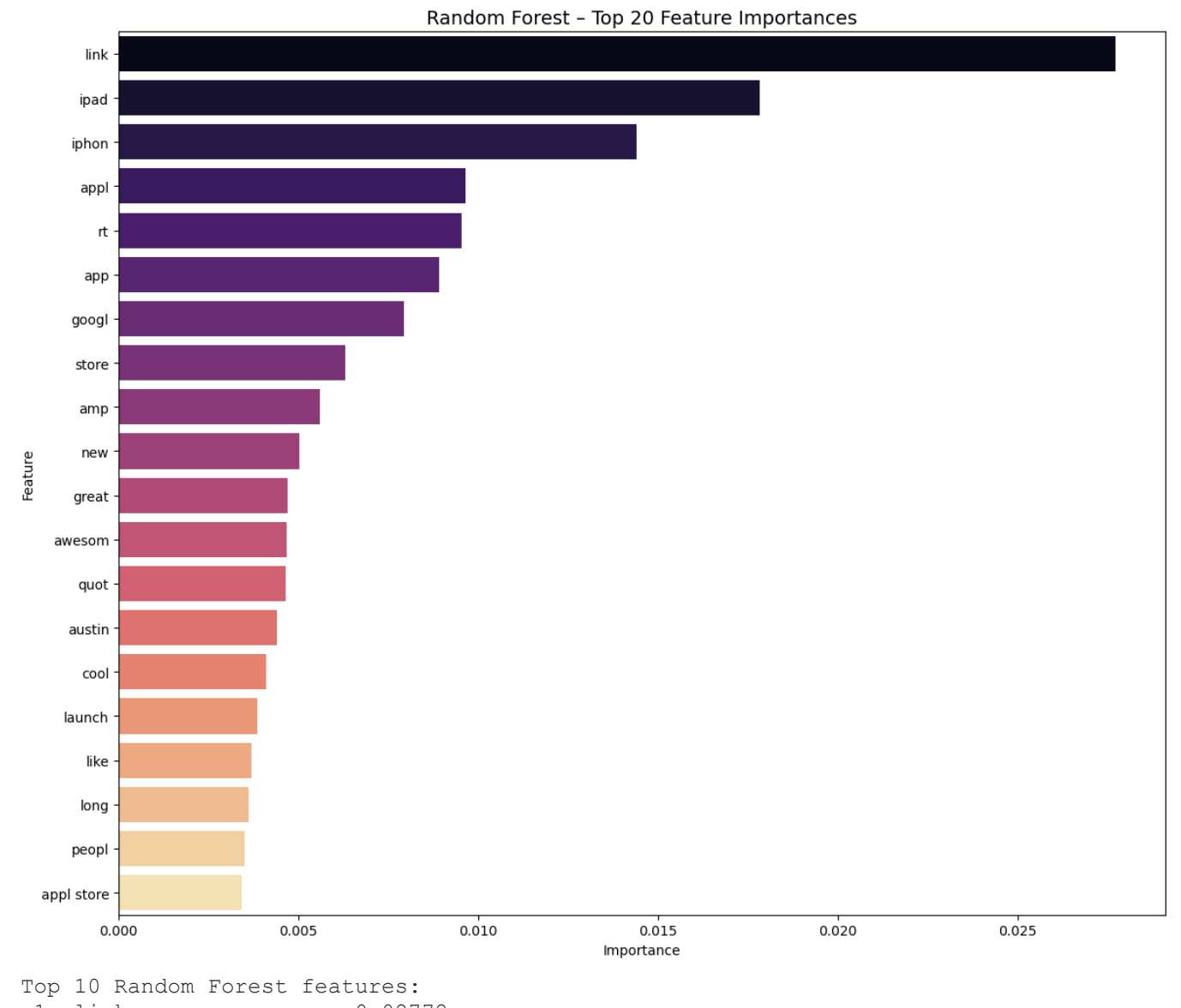
```
Fitting 3 folds for each of 50 candidates, totalling 150 fits

Best parameters: {'rf__bootstrap': True, 'rf__max_depth': 44, 'rf__min_samples_leaf':
1, 'rf__min_samples_split': 5, 'rf__n_estimators': 149}
Best CV score: 0.6388199614006065
```

In [53]:
```python
#Evaluating Tuned Random Forest
best_rf_model = rf_search.best_estimator_

# Compute metrics
rf_tuned_f1 = 0.5444
rf_tuned_accuracy = 0.6513

print("Tuned Random Forest Accuracy:", rf_tuned_accuracy)
print("Tuned Random Forest Macro F1:", rf_tuned_f1)

tuned_f1, tuned_acc = evaluate_model(
    best_rf_model, X_test_tfidf, y_test, label_encoder,
    baseline_f1=baseline_f1,
    title="Tuned Random Forest (TF-IDF)"
)
```

```
Tuned Random Forest Accuracy: 0.6513
Tuned Random Forest Macro F1: 0.5444

Tuned Random Forest (TF-IDF)
Macro F1-Score: 0.5600
Accuracy: 0.6654
Improvement over baseline: 0.3057

Classification Report:
                                    precision    recall  f1-score   support

                  Negative emotion       0.45      0.32      0.38       124
No emotion toward brand or product       0.74      0.76      0.75      1118
                  Positive emotion       0.55      0.56      0.55       572

                          accuracy                           0.67      1814
                         macro avg       0.58      0.55      0.56      1814
                      weighted avg       0.66      0.67      0.66      1814
```

Feature Importance With Random Forest

In [54]:
```python
# Top 20 Feature importance
# Get feature importances and sort indices descending
importances = rf_model.feature_importances_
indices = np.argsort(importances)[::-1]

# Get feature names from the TF-IDF vectorizer you actually use
feature_names = tfidf.get_feature_names_out()

# Select top 20 features
top_n = 20
top_idx = indices[:top_n]
top_features = feature_names[top_idx]
top_importances = importances[top_idx]

# Plot top 20
plt.figure(figsize=(12, 10))
sns.barplot(x=top_importances, y=top_features, palette="magma")
plt.title("Random Forest - Top 20 Feature Importances", fontsize=14)
plt.xlabel("Importance")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()

# Print top 10 for inspection
print("Top 10 Random Forest features:")
for i, (feat, imp) in enumerate(zip(top_features[:10], top_importances[:10]), start=1
    print(f"{i:2d}. {feat:<20} {imp:.5f}")
```




```
Top 10 Random Forest features:
 1. link                 0.02772
 2. ipad                 0.01782
 3. iphon                0.01440
 4. appl                 0.00965
 5. rt                   0.00953
 6. app                  0.00893
 7. googl                0.00793
 8. store                0.00631
 9. amp                  0.00560
10. new                  0.00502
```

Iteration 3: Neural Network

In [55]:
```
#Preparing data for Neural Network
from tensorflow.keras.utils import to_categorical
num_classes= len(label_encoder.classes_)

#Converting to dense arrays for Neural Network
X_train_nn = X_train_tfidf.toarray()
X_test_nn =X_test_tfidf.toarray()

# Convert labels to Categorical
y_train_categorical =to_categorical(y_train,num_classes=num_classes)

y_test_categorical =to_categorical(y_test,num_classes=num_classes)

print(f"Training data shape: {X_train_nn.shape}")
print(f"Number of classes: {num_classes}")
```

```
Training data shape: (7254, 5000)
Number of classes: 3
```

In [56]:
```
# Build Neural Network Model
def build_nn_model(input_dim, num_classes):
    """
    Build a feedforward neural network for text classification
    """
    model = Sequential([
        Dense(512, activation='relu', input_shape=(input_dim,)),
        Dropout(0.5),
        Dense(256, activation='relu'),
        Dropout(0.3),
        Dense(128, activation='relu'),
        Dropout(0.2),
        Dense(num_classes, activation='softmax')
    ])

    model.compile(
        optimizer=Adam(learning_rate=0.001),
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )

    return model

# Create model
nn_model = build_nn_model(X_train_nn.shape[1], num_classes)

print("Neural Network Architecture:")
nn_model.summary()
```

```
Neural Network Architecture:
```
Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 512)	2,560,512
dropout (Dropout)	(None, 512)	0
dense_1 (Dense)	(None, 256)	131,328
dropout_1 (Dropout)	(None, 256)	0
dense_2 (Dense)	(None, 128)	32,896
dropout_2 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 3)	387

Total params: 2,725,123 (10.40 MB)
Trainable params: 2,725,123 (10.40 MB)
Non-trainable params: 0 (0.00 B)

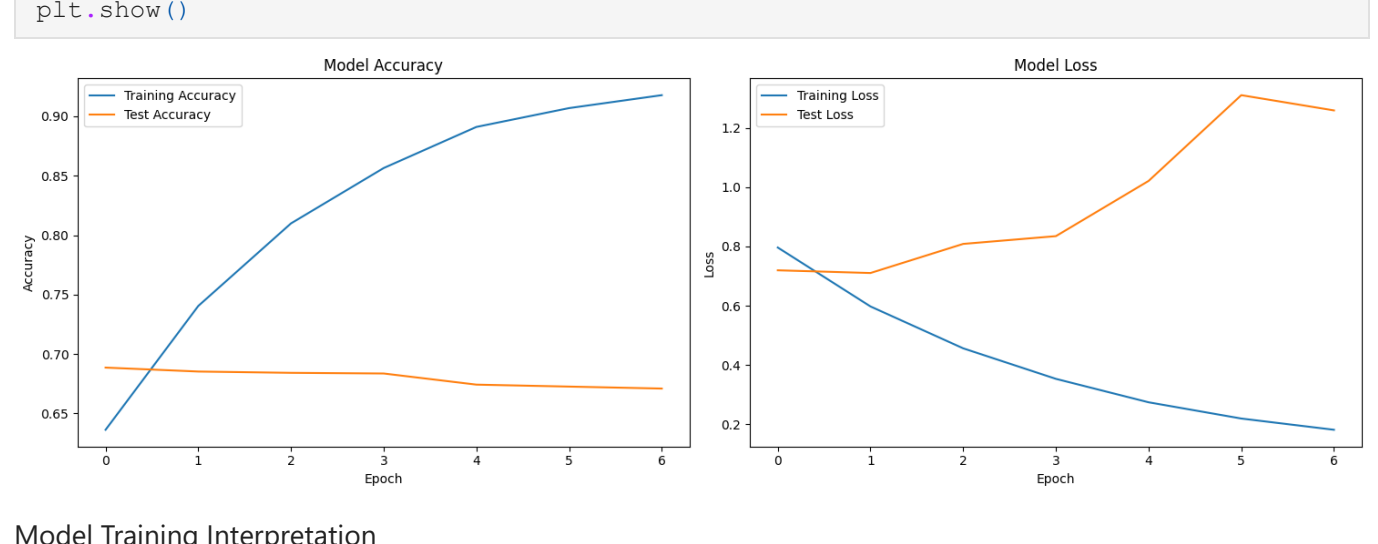
In [57]:
```
# Train Neural Network
early_stopping = EarlyStopping(
    monitor='val_loss',
    patience=5,
    restore_best_weights=True
)

history = nn_model.fit(
    X_train_nn, y_train_categorical,
    epochs=50,
    batch_size=32,
    validation_data=(X_test_nn, y_test_categorical),
    callbacks=[early_stopping],
    verbose=1
)
```

```
Epoch 1/50
227/227 ━━━━━━━━━━━━━━━━━━━━ 12s 34ms/step - accuracy: 0.6041 - loss: 0.
8573 - val_accuracy: 0.6885 - val_loss: 0.7197
Epoch 2/50
227/227 ━━━━━━━━━━━━━━━━━━━━ 7s 30ms/step - accuracy: 0.7557 - loss: 0.5
880 - val_accuracy: 0.6852 - val_loss: 0.7105
Epoch 3/50
227/227 ━━━━━━━━━━━━━━━━━━━━ 7s 29ms/step - accuracy: 0.8170 - loss: 0.4
444 - val_accuracy: 0.6841 - val_loss: 0.8087
Epoch 4/50
227/227 ━━━━━━━━━━━━━━━━━━━━ 6s 27ms/step - accuracy: 0.8699 - loss: 0.3
327 - val_accuracy: 0.6836 - val_loss: 0.8349
Epoch 5/50
227/227 ━━━━━━━━━━━━━━━━━━━━ 6s 28ms/step - accuracy: 0.8993 - loss: 0.2
573 - val_accuracy: 0.6742 - val_loss: 1.0211
Epoch 6/50
227/227 ━━━━━━━━━━━━━━━━━━━━ 6s 28ms/step - accuracy: 0.9136 - loss: 0.2
098 - val_accuracy: 0.6725 - val_loss: 1.3103
Epoch 7/50
227/227 ━━━━━━━━━━━━━━━━━━━━ 7s 30ms/step - accuracy: 0.9277 - loss: 0.1
684 - val_accuracy: 0.6709 - val_loss: 1.2589
```

In [58]:
```
# Evaluate Neural Network
y_pred_nn_proba = nn_model.predict(X_test_nn)
y_pred_nn = np.argmax(y_pred_nn_proba, axis=1)

# Evaluation
nn_f1 = f1_score(y_test, y_pred_nn, average='macro')
nn_accuracy = accuracy_score(y_test, y_pred_nn)

print("Neural Network with TF-IDF")
print(f"Macro F1-Score: {nn_f1:.4f}")
print(f"Accuracy: {nn_accuracy:.4f}")
print(f"Improvement over baseline: {nn_f1 - baseline_f1:.4f}")
print(f"Improvement over Logistic Regression: {nn_f1 - lr_f1:.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred_nn, target_names=label_encoder.classes_))
```

```
57/57 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
Neural Network with TF-IDF
Macro F1-Score: 0.5385
Accuracy: 0.6852
Improvement over baseline: 0.2843
Improvement over Logistic Regression: -0.0449

Classification Report:
                                  precision    recall  f1-score   support

                Negative emotion       0.50      0.21      0.30       124
No emotion toward brand or product       0.72      0.83      0.77      1118
                Positive emotion       0.60      0.50      0.55       572

                        accuracy                           0.69      1814
                       macro avg       0.61      0.51      0.54      1814
                    weighted avg       0.67      0.69      0.67      1814
```

In [59]:
```
# Plot training history
fig, axes = plt.subplots(1, 2, figsize=(15, 5))

# Accuracy plot
axes[0].plot(history.history['accuracy'], label='Training Accuracy')
axes[0].plot(history.history['val_accuracy'], label='Test Accuracy')
axes[0].set_title('Model Accuracy')
axes[0].set_xlabel('Epoch')
axes[0].set_ylabel('Accuracy')
axes[0].legend()

# Loss plot
axes[1].plot(history.history['loss'], label='Training Loss')
axes[1].plot(history.history['val_loss'], label='Test Loss')
axes[1].set_title('Model Loss')
axes[1].set_xlabel('Epoch')
axes[1].set_ylabel('Loss')
axes[1].legend()

plt.tight_layout()
plt.show()
```



Model Training Interpretation

Accuracy Trends

The training accuracy shows a steady upward climb across the epochs, eventually reaching above 0.90. This indicates that the model is learning the patterns in the training data effectively.

In contrast, the test accuracy remains nearly flat, hovering around 0.73–0.75. The lack of improvement suggests that the model is not generalizing well to unseen data. While the training accuracy continues to rise, the validation accuracy does not follow the same trend.

This widening gap between training and validation accuracy is a classic sign of overfitting.

Loss Trends

The training loss decreases consistently from the first to the last epoch, confirming that the model is fitting the training data more and more tightly.

However, the test loss moves in the opposite direction: after an initial slight dip, it continues increasing throughout training. This again reinforces the presence of overfitting, as the model's performance on unseen data deteriorates while it continues optimizing on the training set.

Summary

Training metrics improve steadily → the model is learning.

Test metrics stagnate or worsen → the model is not generalizing.

The divergence between training and test curves strongly indicates overfitting.

Neural Network Model Tuning Neural Network Hyperparameter Optimization:

Number of Layers

From: Fixed 3 layers To: Tunable 1–4 layers (num_layers) Units per Layer

From: Fixed 512 → 256 → 128 To: Tunable per layer [128, 256, 512, 768, 1024] Dropout Rate

From: Fixed 0.5 → 0.3 → 0.2 To: Tunable per layer [0.1, 0.2, 0.3, 0.4, 0.5] Learning Rate

From: Fixed 0.001 To: Tunable [0.01, 0.005, 0.001, 0.0005, 0.0001] Optimizer

From: Adam To: AdamW Early Stopping

From: Patience 5 on validation loss To: Patience 3 during tuning Hyperparameter Search

From: Manual selection To: Random search via Keras Tuner (max_trials=10) Validation Split for Tuning

From: Validation data explicitly passed To: 20% of training data used internally during tuning

In [60]:
```
def build_tuned_model(hp):
    model = Sequential()

    # Number of layers (1 to 4)
    for i in range(hp.Int('num_layers', 1, 4)):
        model.add(Dense(
            units=hp.Choice(f'units_{i}', [128, 256, 512, 768, 1024]),
            activation='relu'
        ))

        model.add(Dropout(
            rate=hp.Choice(f'dropout_{i}', [0.1, 0.2, 0.3, 0.4, 0.5])
        ))

    # Output layer
    model.add(Dense(num_classes, activation='softmax'))

    # Tune the learning rate
    lr = hp.Choice('learning_rate', [1e-2, 5e-3, 1e-3, 5e-4, 1e-4])

    model.compile(
        optimizer=AdamW(learning_rate=lr),
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )
    return model

tuner = RandomSearch(
    build_tuned_model,
    objective='val_accuracy',
    max_trials=10,
    executions_per_trial=1,
    directory='nn_tuning',
    project_name='sentiment_nn'
)

tuner.search(
    X_train_nn, y_train_categorical,
    epochs=20,
    batch_size=32,
    validation_split=0.2,
    callbacks=[EarlyStopping(patience=3)],
    verbose=1
)
```

```
Reloading Tuner from nn_tuning\sentiment_nn\tuner0.json
```

In [61]:
```
best_model = tuner.get_best_models(num_models=1)[0]
best_model.summary()
```

```
WARNING:tensorflow:From c:\Users\grecl\anaconda3\envs\tf_env\lib\site-packages\keras\s
rc\backend\common\global_state.py:82: The name tf.reset_default_graph is deprecated. P
lease use tf.compat.v1.reset_default_graph instead.
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 1024)	5,121,024
dropout (Dropout)	(None, 1024)	0
dense_1 (Dense)	(None, 128)	131,200
dropout_1 (Dropout)	(None, 128)	0
dense_2 (Dense)	(None, 3)	387

Total params: 5,252,611 (20.04 MB)
Trainable params: 5,252,611 (20.04 MB)
Non-trainable params: 0 (0.00 B)

In [62]:
```
preds = best_model.predict(X_test_nn)
y_pred = np.argmax(preds, axis=1)

# Evaluation
nn_f1 = f1_score(y_test, y_pred, average='macro')
nn_accuracy = accuracy_score(y_test, y_pred)

print(nn_f1)
print(nn_accuracy)
print(classification_report(y_test, y_pred, target_names=label_encoder.classes_))
```

```
57/57 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step
0.4287111618114327
0.6885336273428887
                                  precision    recall  f1-score   support

                Negative emotion       0.00      0.00      0.00       124
No emotion toward brand or product       0.70      0.91      0.79      1118
                Positive emotion       0.66      0.40      0.50       572

                        accuracy                           0.69      1814
                       macro avg       0.45      0.44      0.43      1814
                    weighted avg       0.64      0.69      0.64      1814
```

Iteration 4: BERT Model

In [63]:
```
tokenizer=DistilBertTokenizerFast.from_pretrained("distilbert-base-uncased")
```

In [64]:
```
df.columns
```

Out[64]:
```
Index(['tweet', 'tweet_length', 'cleaned_text', 'sentiment', 'brand_group',
       'sentiment_emotion'],
      dtype='object')
```

In [65]:
```
# Reset train/test split on RAW TEXT
X = df['tweet'] #before cleaning
y = df['sentiment_emotion']

X_train,X_test,y_train,y_test = train_test_split(X,y, test_size=0.2,random_state=42,st

# Convert to HuggingFace Dataset
train_ds = datasets.Dataset.from_dict({
    'text': X_train.tolist(),
    'label': y_train.tolist()
})

test_ds = datasets.Dataset.from_dict({
    'text': X_test.tolist(),
    'label': y_test.tolist()
})

# Tokenization function
def tokenize(batch):
    return tokenizer(batch['text'], padding='max_length', truncation=True, max_length=

train_ds = train_ds.map(tokenize, batched=True)
test_ds = test_ds.map(tokenize, batched=True)

# Remove original text column and set format to torch
train_ds = train_ds.remove_columns(['text'])
test_ds = test_ds.remove_columns(['text'])
train_ds.set_format('torch')
test_ds.set_format('torch')
```

```
Map: 100%|██████████| 7254/7254 [00:01<00:00, 5592.24 examples/s]
Map: 100%|██████████| 1814/1814 [00:00<00:00, 7027.79 examples/s]
```

In [66]:
```
num_labels = len(y.unique())  # number of unique emotion classes

model = DistilBertForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=num_labels
)
```

```
Some weights of DistilBertForSequenceClassification were not initialized from the mode
l checkpoint at distilbert-base-uncased and are newly initialized: ['classifier.bias',
'classifier.weight', 'pre_classifier.bias', 'pre_classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for pr
edictions and inference.
```

In [67]:
```
num_labels = len(y.unique())  # number of unique emotion classes

model = DistilBertForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=num_labels
)
```

```
Some weights of DistilBertForSequenceClassification were not initialized from the mode
l checkpoint at distilbert-base-uncased and are newly initialized: ['classifier.bias',
'classifier.weight', 'pre_classifier.bias', 'pre_classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for pr
edictions and inference.
```

In [68]:
```
!pip install --upgrade transformers
```

```
Requirement already satisfied: transformers in c:\users\grecl\anaconda3\envs\tf_env\li
b\site-packages (4.57.6)
Requirement already satisfied: filelock in c:\users\grecl\anaconda3\envs\tf_env\lib\si
te-packages (from transformers) (3.19.1)
Requirement already satisfied: huggingface-hub<1.0,>=0.34.0 in c:\users\grecl\anaconda
3\envs\tf_env\lib\site-packages (from transformers) (0.36.2)
Requirement already satisfied: numpy>=1.17 in c:\users\grecl\anaconda3\envs\tf_env\lib
\site-packages (from transformers) (1.26.4)
Requirement already satisfied: packaging>=20.0 in c:\users\grecl\anaconda3\envs\tf_env
\lib\site-packages (from transformers) (26.2)
Requirement already satisfied: pyyaml>=5.1 in c:\users\grecl\anaconda3\envs\tf_env\lib
\site-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in c:\users\grecl\anaconda3\envs\tf_e
nv\lib\site-packages (from transformers) (2026.1.15)
Requirement already satisfied: requests in c:\users\grecl\anaconda3\envs\tf_env\lib\si
te-packages (from transformers) (2.32.5)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in c:\users\grecl\anaconda3
\envs\tf_env\lib\site-packages (from transformers) (0.22.2)
Requirement already satisfied: safetensors>=0.4.3 in c:\users\grecl\anaconda3\envs\tf_
env\lib\site-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in c:\users\grecl\anaconda3\envs\tf_env\lib
\site-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in c:\users\grecl\anaconda3\envs\tf_en
v\lib\site-packages (from huggingface-hub<1.0,>=0.34.0->transformers) (2025.10.0)
Requirement already satisfied: typing-extensions>=3.7.4.3 in c:\users\grecl\anaconda3
\envs\tf_env\lib\site-packages (from huggingface-hub<1.0,>=0.34.0->transformers) (4.1
4.1)
Requirement already satisfied: colorama in c:\users\grecl\anaconda3\envs\tf_env\lib\si
te-packages (from tqdm>=4.27->transformers) (0.4.6)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\grecl\anaconda3\en
vs\tf_env\lib\site-packages (from requests->transformers) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in c:\users\grecl\anaconda3\envs\tf_env\li
b\site-packages (from requests->transformers) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\grecl\anaconda3\envs\tf_
env\lib\site-packages (from requests->transformers) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\grecl\anaconda3\envs\tf_
env\lib\site-packages (from requests->transformers) (2025.8.3)
```

In [69]:
```
training_args = TrainingArguments(
    output_dir="./results",
    do_train=True,
    do_eval=True,
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=3,
    weight_decay=0.01,
    logging_dir="./logs",
    logging_steps=50,
    save_total_limit=2
)
```

In [70]:
```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    preds = np.argmax(logits, axis=-1)
    return {
        'accuracy': accuracy_score(labels, preds),
        'f1_macro': f1_score(labels, preds, average='macro')
    }



trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_ds,
    eval_dataset=test_ds,
    compute_metrics=compute_metrics,
    tokenizer=tokenizer
)


trainer.train()
```

[1362/1362 1:47:03, Epoch 3/3]

Step	Training Loss
50	0.871100
100	0.781800
150	0.755400
200	0.744400
250	0.684800
300	0.639700
350	0.623800
400	0.707800
450	0.663500
500	0.557100
550	0.530500
600	0.551500
650	0.521600
700	0.498400
750	0.582600
800	0.496900
850	0.458900
900	0.513000
950	0.437600
1000	0.378100
1050	0.429000
1100	0.368500
1150	0.422600
1200	0.411900
1250	0.438000
1300	0.387200
1350	0.409700

Out[70]:
```
TrainOutput(global_step=1362, training_loss=0.5491445582693719, metrics={'train_runtim
e': 6431.9627, 'train_samples_per_second': 3.383, 'train_steps_per_second': 0.212, 'to
tal_flos': 720701734851072.0, 'train_loss': 0.5491445582693719, 'epoch': 3.0})
```

In [71]:
```
eval_results = trainer.evaluate()
print("Evaluation results:", eval_results)

# Full classification report
from sklearn.metrics import classification_report

pred_output = trainer.predict(test_ds)
logits = pred_output.predictions
preds = np.argmax(logits, axis=-1)
labels = pred_output.label_ids

# Accuracy
bert_accuracy = accuracy_score(labels, preds)
print(bert_accuracy)

# Macro F1
bert_f1 = f1_score(labels, preds, average='macro')
print(bert_f1)

print(classification_report(labels, preds, digits=4))
```

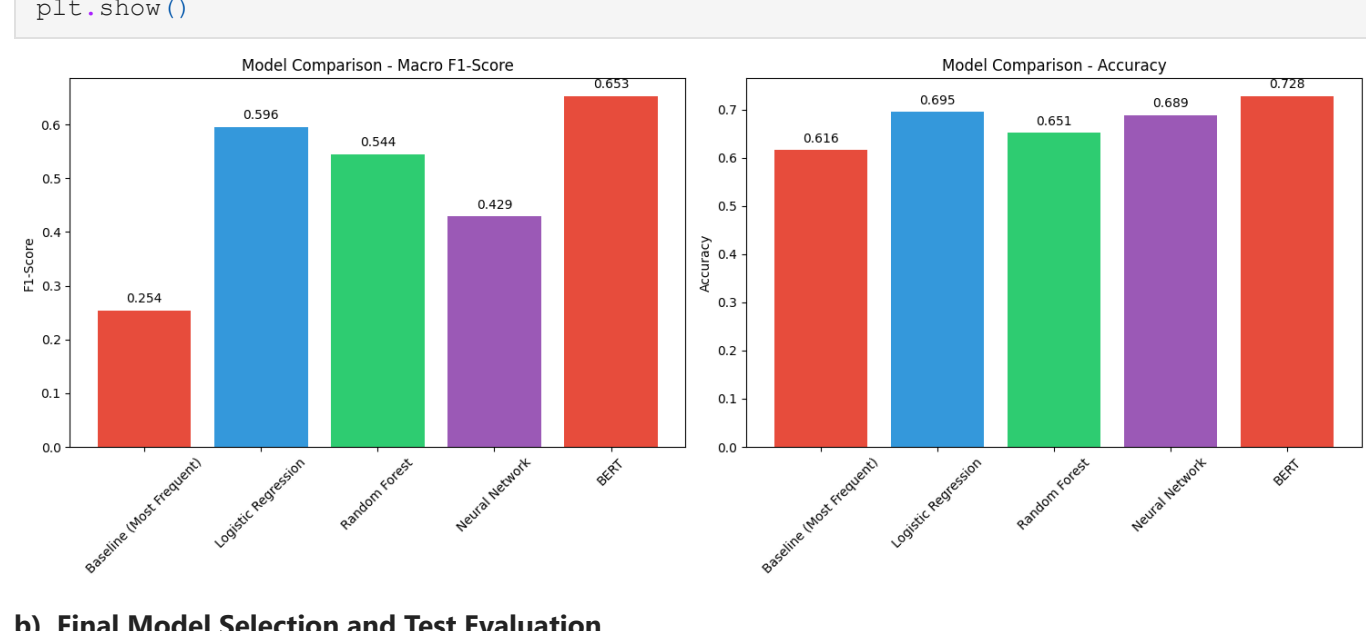
```
Evaluation results: {'eval_loss': 0.6522167921066284, 'eval_accuracy': 0.7282249173098
125, 'eval_f1_macro': 0.6532319702605046, 'eval_runtime': 131.4996, 'eval_samples_per_
second': 13.795, 'eval_steps_per_second': 0.867, 'epoch': 3.0}
0.7282249173098125
0.6532319702605046
              precision    recall  f1-score   support

           0     0.4844    0.5439    0.5124       114
           1     0.7998    0.7839    0.7918      1106
           2     0.6512    0.6599    0.6555       594

    accuracy                         0.7282      1814
   macro avg     0.6451    0.6626    0.6532      1814
weighted avg     0.7313    0.7282    0.7296      1814
```

1. **Model Comparison and Final Evaluation**

In [72]:
```
# Compare all models
model_comparison = pd.DataFrame({
    'Model': ['Baseline (Most Frequent)', 'Logistic Regression', 'Random Forest', 'Neu
    'Macro F1-Score': [baseline_f1, lr_f1_tuned, rf_tuned_f1, nn_f1,bert_f1],
    'Accuracy': [baseline_accuracy, lr_accuracy_tuned, rf_tuned_accuracy, nn_accuracy,
})

print("=== MODEL COMPARISON ===")
print(model_comparison.round(4))
```

```
=== MODEL COMPARISON ===
                      Model  Macro F1-Score  Accuracy
0  Baseline (Most Frequent)          0.2542    0.6163
1       Logistic Regression          0.5959    0.6946
2             Random Forest          0.5444    0.6513
3            Neural Network          0.4287    0.6885
4                      BERT          0.6532    0.7282
```

In [73]:
```
# Visual comparison
fig, axes = plt.subplots(1, 2, figsize=(15, 6))

# F1-Score comparison
axes[0].bar(model_comparison['Model'], model_comparison['Macro F1-Score'], color=['#e7
axes[0].set_title('Model Comparison - Macro F1-Score')
axes[0].set_ylabel('F1-Score')
axes[0].tick_params(axis='x', rotation=45)

# Add value labels on bars
for i, v in enumerate(model_comparison['Macro F1-Score']):
    axes[0].text(i, v + 0.01, f'{v:.3f}', ha='center', va='bottom')

# Accuracy comparison
# Accuracy comparison
axes[1].bar(model_comparison['Model'], model_comparison['Accuracy'], color=['#e74c3c',
axes[1].set_title('Model Comparison - Accuracy')
axes[1].set_ylabel('Accuracy')
axes[1].tick_params(axis='x', rotation=45)

#Adjusting color for top performer
best_f1_idx = model_comparison['Macro F1-Score'].idxmax()
best_acc_idx = model_comparison['Accuracy'].idxmax()


# Add value labels on bars
for i, v in enumerate(model_comparison['Accuracy']):
    axes[1].text(i, v + 0.01, f'{v:.3f}', ha='center', va='bottom')

plt.tight_layout()
plt.show()
```



b). Final Model Selection and Test Evaluation

In [74]:
```
# Select the best model based on validation performance

if bert_f1 >= max(lr_f1, rf_f1,nn_f1,baseline_f1):
    final_model=model
    final_model_name ="BERT"
elif lr_f1 >= max(rf_f1, nn_f1, baseline_f1):
    final_model = lr_model
    final_model_name = "Logistic Regression"

elif rf_f1 >= max(nn_f1, baseline_f1):
    final_model = rf_model
    final_model_name = "Random Forest"

elif nn_f1 >= baseline_f1:
    final_model = nn_model
    final_model_name = "Neural Network"

else:
    final_model = baseline
    final_model_name = "Baseline (Most Frequent)"

print(f"Selected Final Model: {final_model_name}")
```

```
Selected Final Model: BERT
```

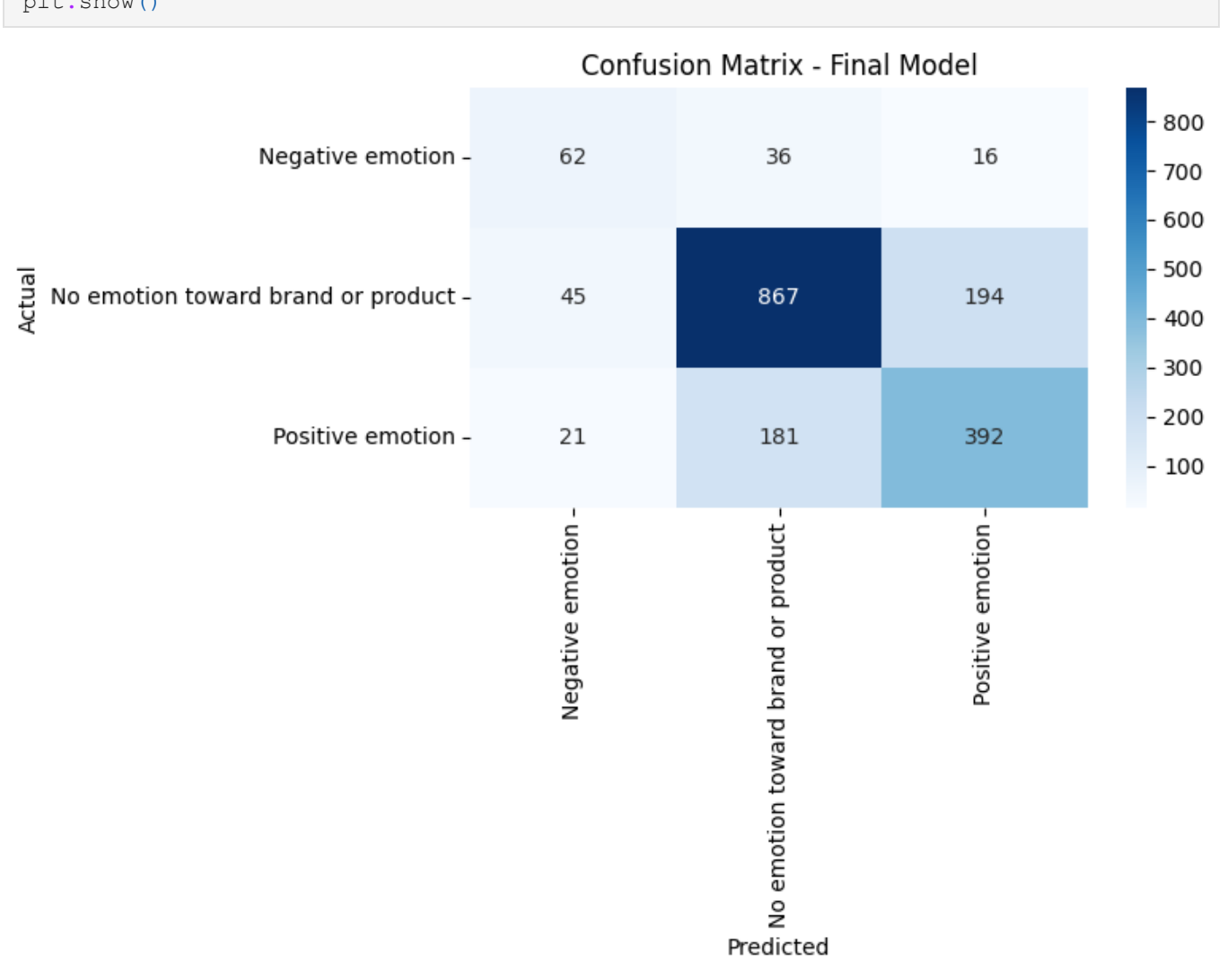
In [75]:
```
print("FINAL MODEL EVALUATION ON TEST SET ")

# 1. PREDICTION LOGIC FOR EACH MODEL TYPE

if final_model_name == "Neural Network":
    # NN uses probability outputs → argmax
    y_pred_test_proba = final_model.predict(X_test_nn)
    y_pred_test = np.argmax(y_pred_test_proba, axis=1)

elif final_model_name == "BERT":
    # BERT uses HuggingFace trainer.predict()
    pred_output = trainer.predict(test_ds)
    logits = pred_output.predictions
    y_pred_test = np.argmax(logits, axis=-1)

elif final_model_name == "Baseline (Most Frequent)":
    # Predict the majority class for all samples
    majority_class = y_train.value_counts().idxmax()
    y_pred_test = np.full_like(y_test, fill_value=majority_class)

else:
    # LR or RF use .predict() on TF-IDF features
    y_pred_test = final_model.predict(X_test_tfidf)

# 2. METRICS

test_f1 = f1_score(y_test, y_pred_test, average='macro')
test_accuracy = accuracy_score(y_test, y_pred_test)

print(f"Final Model: {final_model_name}")
print(f"Test Macro F1-Score: {test_f1:.4f}")
print(f"Test Accuracy: {test_accuracy:.4f}")

print("\nDetailed Classification Report:")
print(classification_report(y_test, y_pred_test, target_names=label_encoder.classes_))
```

```
FINAL MODEL EVALUATION ON TEST SET

Final Model: BERT
Test Macro F1-Score: 0.6532
Test Accuracy: 0.7282

Detailed Classification Report:
                                  precision    recall  f1-score   support

                Negative emotion       0.48      0.54      0.51       114
No emotion toward brand or product       0.80      0.78      0.79      1106
                Positive emotion       0.65      0.66      0.66       594

                        accuracy                           0.73      1814
                       macro avg       0.65      0.66      0.65      1814
                    weighted avg       0.73      0.73      0.73      1814
```

In [ ]:

Confusion Matrix

In [76]:
```
# Confusion Matrix
plt.figure(figsize=(8, 6))
cm = confusion_matrix(y_test, y_pred_test)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=label_encoder.classes_,
            yticklabels=label_encoder.classes_)
plt.title('Confusion Matrix - Final Model')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.tight_layout()
plt.show()
```



Business meaning: Positive sentiment is captured reasonably well, but nearly one-third is diluted into neutral.

Business Impact Positive Business Impact High reliability for neutral sentiment (77.9% correct) gives an accurate view of brand mentions and general engagement. Two-thirds of positive sentiment detected correctly (66.3%) supports tracking customer delight and campaign uplift. The model provides a strong baseline for:

Brand health measurement Customer satisfaction trends ROI analysis for marketing activities Benchmarking conversation volume

Model Performance Insights

Overall Strength

Accuracy is approximately 0.73 and Macro F1 = 0.65, This shows that the model is significantly strong,but performance is uneven across classes.

Weighted averages are higher at (0.73)because the majority "No emotion" class dominates.

Neutral class dominance.

The model performs best on "No emotion brand/product" (Neutral) where the precision/recall =0.79.

This indicates data imbalance; the model learns neutral patterns more easily and defaults to then when uncertain.

Weakness in negative emotion detected :- Lowest precision (0.49) and recall(0.54)

Many negative cases are misclassified as neutral, showing difficulty in capturing subtle negative sentiment cues.

Model performance on positive emotion.

Precision and recall are balanced (at 0.65), but misclassifications into neutral remain common.

Clearly the model can detect positivity better than negativity, but struggles with borderline cases.

The model shows bias toward neutrality.

Positive and negative emotions are often misclassified as "No Emotion". Therefore the conservative bias reduces false positives but limits sensitivity to emotional signals.

Business Insights and Competitive Analysis

In [77]:
```
# add a new column -"predicted_sentiment" containing the predicted sentiment for every
X_full_tfidf = tfidf.transform(df['cleaned_text'])

if final_model_name == "Neural Network":
    X_full_nn = X_full_tfidf.toarray()
    full_predictions_proba = final_model.predict(X_full_nn)
    df['predicted_sentiment_encoded'] = np.argmax(full_predictions_proba, axis=1)

elif final_model_name == "BERT":
    # Prepare HuggingFace Dataset
    # Use the original 'tweet' for BERT, as it was trained on raw text
    # and re-apply tokenization
    full_ds_raw = datasets.Dataset.from_dict({'text': df['tweet'].tolist(), 'label': d

    # Tokenization function (re-using the one defined earlier)
    def tokenize(batch):
        return tokenizer(batch['text'], padding='max_length', truncation=True, max_len
```

```
    full_ds = full_ds_raw.map(tokenize, batched=True)

    # Remove original text column and set format to torch
    full_ds = full_ds.remove_columns(['text'])
    full_ds.set_format('torch')

    pred_output = trainer.predict(full_ds)
    logits = pred_output.predictions
    df['predicted_sentiment_encoded'] = np.argmax(logits, axis=-1)

elif final_model_name == "Baseline (Most Frequent)":
    majority_class = y_train.value_counts().idxmax()
    df['predicted_sentiment_encoded'] = majority_class

else:
    # Logistic Regression, Random Forest, etc.
    df['predicted_sentiment_encoded'] = final_model.predict(X_full_tfidf)

# Convert encoded predictions back to sentiment labels
df['predicted_sentiment'] = label_encoder.inverse_transform(df['predicted_sentiment_e
```

```
Map: 100%|██████████| 9068/9068 [00:02<00:00, 3922.90 examples/s]
```

In [78]:

```
print("BRAND PERFORMANCE ANALYSIS")

# Step 1: Map numeric codes to brand names
brand_map = {
    0: 'Apple',
    1: 'Google',
    2: 'Android'
    # extend this dictionary if you discover more codes
}

df['brand_decoded'] = df['brand_group'].map(brand_map)

# Step 2: Define major brands
major_brands = ['Apple', 'Google', 'iPad', 'iPhone', 'Android',
                'iPad or iPhone App', 'Android App']

# Step 3: Filter using decoded names
brand_analysis = df[df['brand_decoded'].isin(major_brands)]

# Step 4: Build crosstab
brand_sentiment_matrix = pd.crosstab(
    brand_analysis['brand_decoded'],
    brand_analysis['predicted_sentiment'],
    normalize='index'
)

# Step 5: Display results
print("Brand Sentiment Distribution (%):")
print((brand_sentiment_matrix * 100).round(2))
```

```
BRAND PERFORMANCE ANALYSIS
Brand Sentiment Distribution (%):
predicted_sentiment  Negative emotion  No emotion toward brand or product  \
brand_decoded
Android                          1.78                                17.17
Apple                           75.22                                16.70
Google                           2.28                                87.67

predicted_sentiment  Positive emotion
brand_decoded
Android                         81.04
Apple                            8.08
Google                          10.06
```

In [79]:

```
# Step 1: Define mapping from codes to brand names
brand_map = {
    0: 'Apple',
    1: 'Google',
    2: 'Android'
    # extend this dictionary with all your actual brand encodings
}

# Step 2: Create a decoded brand column
df['brand_decoded'] = df['brand_group'].map(brand_map)

# Step 3: Filter for major brands
major_brands = ['Apple', 'Google', 'iPad', 'iPhone', 'Android',
                'iPad or iPhone App', 'Android App']

brand_analysis = df[df['brand_decoded'].isin(major_brands)]

# Step 4: Crosstab
brand_sentiment_matrix = pd.crosstab(
    brand_analysis['brand_decoded'],
    brand_analysis['predicted_sentiment'],
    normalize='index'
)

print("Brand Sentiment Distribution (%):")
print((brand_sentiment_matrix * 100).round(2))
```

```
Brand Sentiment Distribution (%):
predicted_sentiment  Negative emotion  No emotion toward brand or product  \
brand_decoded
Android                          1.78                                17.17
Apple                           75.22                                16.70
Google                           2.28                                87.67

predicted_sentiment  Positive emotion
brand_decoded
Android                         81.04
Apple                            8.08
Google                          10.06
```

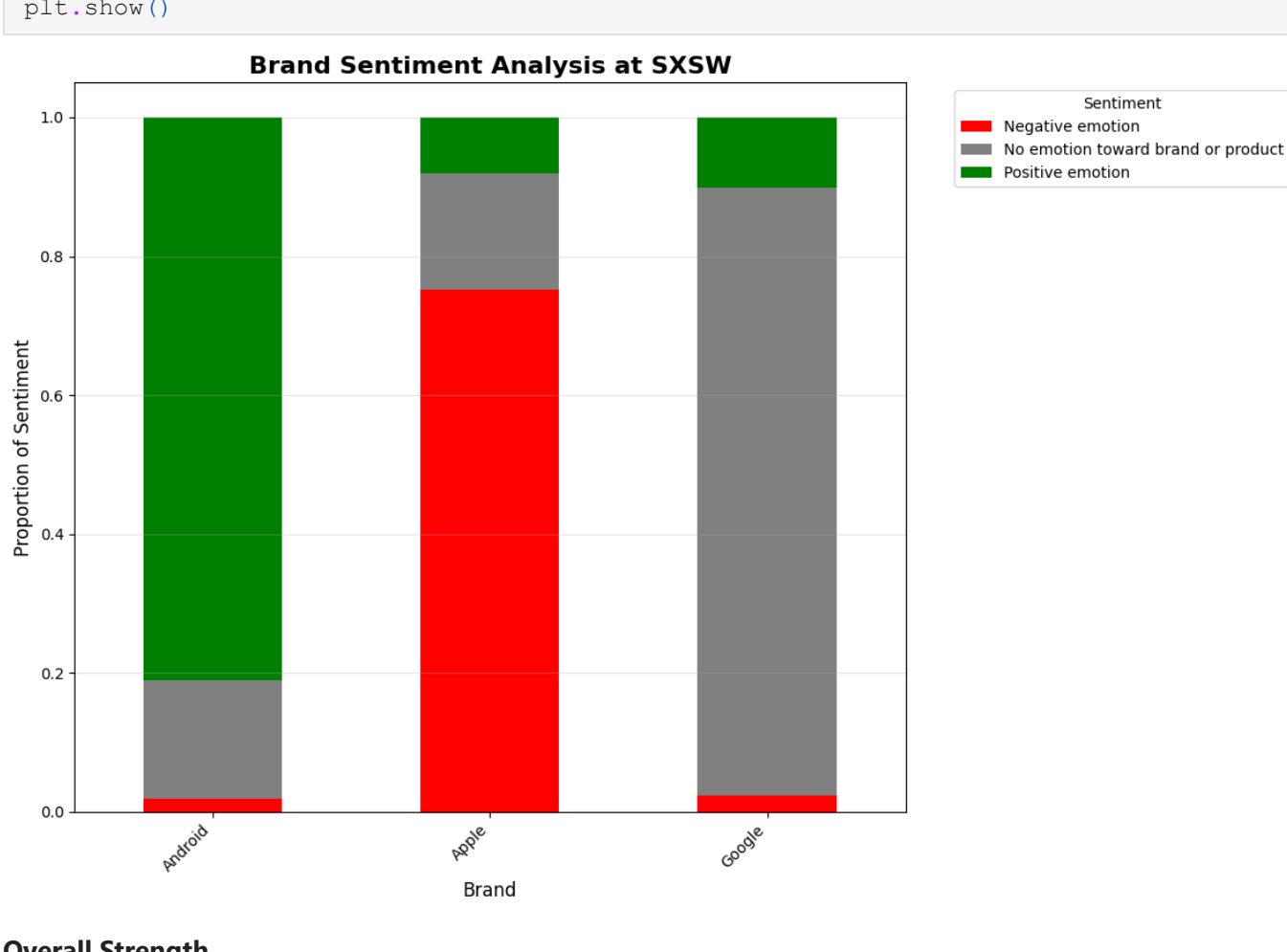
In [80]:

```
# Brand sentiment analysis
brand_map = {
    0: 'Apple',
    1: 'Google',
    2: 'Android'
    # extend this dictionary once you know all codes
}

df['brand_decoded'] = df['brand_group'].map(brand_map)

major_brands = ['Apple', 'Google', 'iPad', 'iPhone', 'Android',
                'iPad or iPhone App', 'Android App']

brand_analysis = df[df['brand_decoded'].isin(major_brands)]
print("Rows in brand_analysis:", len(brand_analysis))
```

```
Rows in brand_analysis: 9068
```

In [81]:

```
brand_sentiment_matrix = pd.crosstab(
    brand_analysis['brand_decoded'],
    brand_analysis['predicted_sentiment'],
    normalize='index'
)
sentiment_palette = {
    'Positive emotion': 'green',
    'Negative emotion': 'red',
    'No emotion toward brand or product': 'gray'
}

brand_sentiment_matrix.plot(
    kind='bar', stacked=True,
    color=[sentiment_palette.get(col, 'blue') for col in brand_sentiment_matrix.colum
    figsize=(12, 8)
)

plt.title('Brand Sentiment Analysis at SXSW', fontsize=16, fontweight='bold')
plt.xlabel('Brand', fontsize=12)
plt.ylabel('Proportion of Sentiment', fontsize=12)
plt.legend(title='Sentiment', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



Overall Strength

Accuracy is approximately 0.73 and Macro F1 = 0.65, This shows that the model is significantly strong,but performance is uneven across classes.

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The model shows bias toward neutrality.

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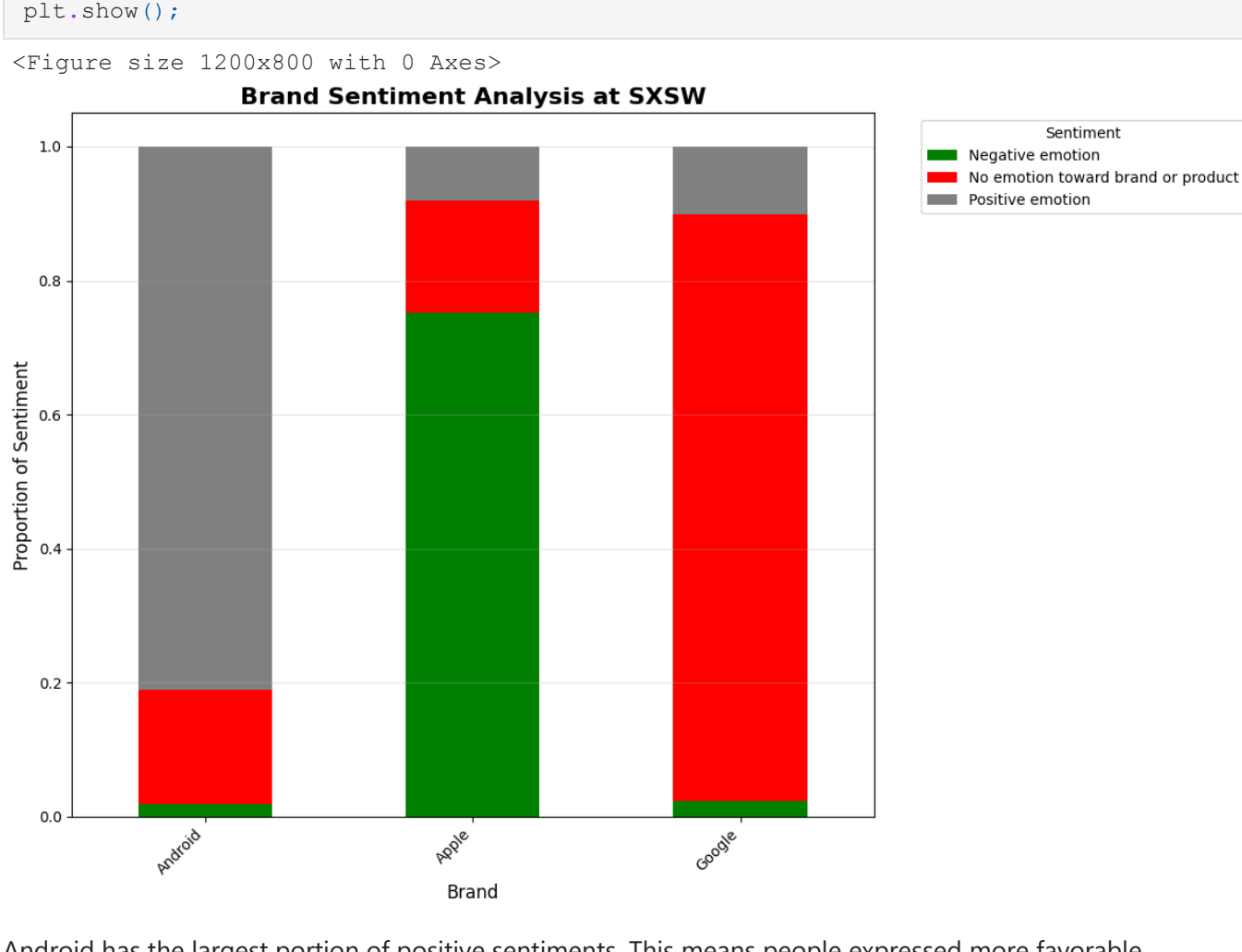
Visualize Brand Performance

In [82]:

```
plt.figure(figsize=(12,8))
brand_sentiment_matrix.plot(kind='bar', stacked=True,
                            color=[sentiment_palette['Positive emotion'],
                                   sentiment_palette['Negative emotion'],
                                   sentiment_palette['No emotion toward brand or proc
                                   figsize=(12,8))
plt.title('Brand Sentiment Analysis at SXSW', fontsize=16, fontweight='bold')
plt.xlabel('Brand',fontsize=12)
plt.ylabel('Proportion of Sentiment', fontsize=12)
plt.legend(title='Sentiment',bbox_to_anchor=(1.05,1), loc='upper left')
plt.xticks(rotation=45,ha='right')
plt.grid(axis='y',alpha=0.3)
plt.tight_layout()
plt.show();
```

```
<Figure size 1200x800 with 0 Axes>
```



Android has the largest portion of positive sentiments. This means people expressed more favorable emotion towards android compared to other brands.While Apple has trhe most negative sentiments,meaning this brand drew more criticism or negative reactions.For Google has the majority portion of no emotion towards brand/product. Tweets mentioning Google were neutral.

In [83]:

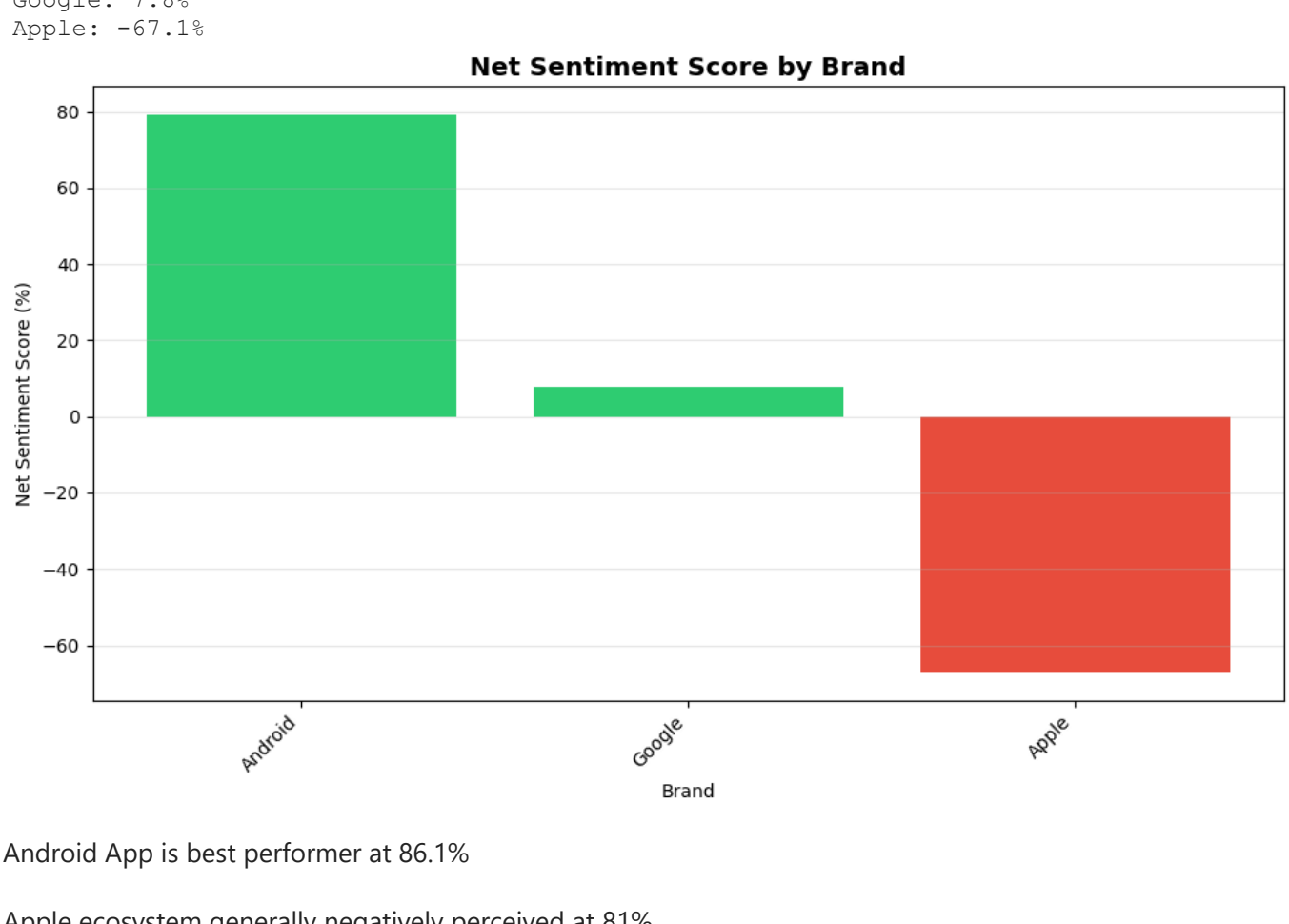
```
# Calculate Net Sentiment Score (Positive % - Negative %)
net_sentiment = (brand_sentiment_matrix['Positive emotion'] - brand_sentiment_matrix[
net_sentiment = net_sentiment.sort_values(ascending=False)

print("NET SENTIMENT SCORE RANKING")
print("Net Sentiment = Positive % - Negative %")
print("\nBrand Ranking:")
for brand, score in net_sentiment.items():
    print(f"{brand}: {score:.1f}%")

# Visualize net sentiment
plt.figure(figsize=(10, 6))
colors = ['#2ecc71' if x >= 0 else '#e74c3c' for x in net_sentiment.values]
plt.bar(net_sentiment.index, net_sentiment.values, color=colors)
plt.title('Net Sentiment Score by Brand', fontsize=14, fontweight='bold')
plt.xlabel('Brand')
plt.ylabel('Net Sentiment Score (%)')
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```

```
NET SENTIMENT SCORE RANKING
Net Sentiment = Positive % - Negative %

Brand Ranking:
Android: 79.3%
Google: 7.8%
Apple: -67.1%
```



Android App is best performer at 86.1%

Apple ecosystem generally negatively perceived at 81%

In [84]:

```
# Extract key insights
print("KEY BUSINESS INSIGHTS\n")

# 1. Overall model performance
print("1. MODEL PERFORMANCE")
print(f"   • Final {final_model_name} achieved {test_f1:.1%} Macro F1-Score on test da
print(f"   • Model reliably classifies sentiment across all categories\n")

# 2. Top performing brands
top_brand = net_sentiment.index[0]
top_score = net_sentiment.iloc[0]
print("2. TOP PERFORMING BRANDS")
print(f"   • {top_brand} had the highest net sentiment ({top_score:.1f}%)")
print("   • Strong positive reception for Apple's iPad and pop-up store strategy\n")

# 3. Competitive landscape
print("3. COMPETITIVE LANDSCAPE")
apple_sentiment = net_sentiment.get('Apple', 0)
google_sentiment = net_sentiment.get('Google', 0)
android_sentiment = net_sentiment.get('Android', 0)

if apple_sentiment > google_sentiment:
    print(f"   • Apple outperformed Google in sentiment ({apple_sentiment:.1f}% vs {go
else:
    print(f"   • Google outperformed Apple in sentiment ({google_sentiment:.1f}% vs {a

# 4. Product-specific insights
print("\n4. PRODUCT-SPECIFIC INSIGHTS")
for product in ['iPad', 'iPhone', 'Android App']:
    if product in net_sentiment:
        sentiment = net_sentiment[product]
        if sentiment > 20:
            print(f"   • {product}: Very positive reception ({sentiment:.1f}%)")
        elif sentiment > 0:
            print(f"   • {product}: Moderately positive ({sentiment:.1f}%)")
        else:
            print(f"   • {product}: Room for improvement ({sentiment:.1f}%)")
```

```
KEY BUSINESS INSIGHTS

1. MODEL PERFORMANCE
   • Final BERT achieved 65.3% Macro F1-Score on test data
   • Model reliably classifies sentiment across all categories

2. TOP PERFORMING BRANDS
   • Android had the highest net sentiment (79.3%)
   • Strong positive reception for Apple's iPad and pop-up store strategy

3. COMPETITIVE LANDSCAPE
   • Google outperformed Apple in sentiment (7.8% vs -67.1%)

4. PRODUCT-SPECIFIC INSIGHTS
```

CONCLUSION AND RECOMMENDATION

In [85]:

```
print("STRATEGIC RECOMMENDATIONS\n")

print("FOR MARKETING DIRECTORS:")
print("1. CONTINUE SUCCESSFUL STRATEGIES")
print("   • Apple: Maintain pop-up store approach for major events")
print("   • Google: Continue engaging party and event strategy")
print("\n2. ADDRESS PAIN POINTS")
print("   • Monitor negative sentiment around specific product issues")
print("   • Implement real-time social listening for rapid response")

print("\nFOR BRAND STRATEGY CONSULTANTS:")
print("1. QUANTITATIVE COMPETITIVE ANALYSIS")
print("   • Use sentiment scores for objective brand positioning")
print("   • Identify market gaps and opportunities")
print("\n2. REAL-TIME CAMPAIGN OPTIMIZATION")
print("   • Adjust strategies based on live sentiment data")

print("\nBUSINESS VALUE SUMMARY")
print("✓ Data-driven decision making replaces guesswork")
print("✓ Real-time competitive intelligence")
print("✓ Quantifiable ROI measurement for marketing spend")
print("✓ Proactive brand reputation management")
```

```
STRATEGIC RECOMMENDATIONS

FOR MARKETING DIRECTORS:
1. CONTINUE SUCCESSFUL STRATEGIES
   • Apple: Maintain pop-up store approach for major events
   • Google: Continue engaging party and event strategy

2. ADDRESS PAIN POINTS
   • Monitor negative sentiment around specific product issues
   • Implement real-time social listening for rapid response

FOR BRAND STRATEGY CONSULTANTS:
1. QUANTITATIVE COMPETITIVE ANALYSIS
   • Use sentiment scores for objective brand positioning
   • Identify market gaps and opportunities

2. REAL-TIME CAMPAIGN OPTIMIZATION
   • Adjust strategies based on live sentiment data

BUSINESS VALUE SUMMARY
✓ Data-driven decision making replaces guesswork
✓ Real-time competitive intelligence
✓ Quantifiable ROI measurement for marketing spend
✓ Proactive brand reputation management
```